36-671 Final Project

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Research Question

How do regional gasoline prices co-move over time, and what lead-lag relationships appear across regions and between crude oil prices and retail gasoline prices?

1. Preparing Time Series

```
# File paths
gas_file_path <- here("data", "PET_PRI_GND_A_EPM0_PTE_DPGAL_W.xls")
brent_file_path <- here("data", "brent_crude_weekly_1993_present.csv")
```

```
# Load weekly U.S. gasoline prices
us_gas_df <- read_excel(gas_file_path, sheet = "Data 1", skip = 2) |>
  rename(
    Date = 1,
    US_All_Grades = 2
  ) |>
  mutate(
    Date = as.Date(Date),
    US_All_Grades = as.numeric(US_All_Grades)
  ) |>
  arrange(Date)

# Load weekly regional gasoline prices
region_gas_df <- read_excel(gas_file_path, sheet = "Data 2", skip = 2) |>
  rename(
    Date = 1,
    East_Coast = 2,
```

```r
    New_England = 3,
    Central_Atlantic = 4,
    Lower_Atlantic = 5,
    Midwest = 6,
    Gulf_Coast = 7,
    Rocky_Mountain = 8,
    West_Coast = 9,
    West_Coast_ex_CA = 10
  ) |>
  mutate(
    Date = as.Date(Date),
    across(-Date, as.numeric)
  ) |>
  arrange(Date)

# Keep the five regions used in the regional VAR
gas_df <- us_gas_df |>
  left_join(
    region_gas_df |>
      dplyr::select(Date, East_Coast, Midwest, Gulf_Coast, Rocky_Mountain, West_Coast),
    by = "Date"
  ) |>
  drop_na()

cat("Weekly gasoline date range:", min(gas_df$Date), "to", max(gas_df$Date), "\n")
```

```
Weekly gasoline date range: 8495 to 20542
```

```r
cat("Gasoline observations:", nrow(gas_df), "\n")
```

```
Gasoline observations: 1722
```

```r
# Load weekly Brent crude oil prices
brent_df <- read_csv(brent_file_path, show_col_types = FALSE) |>
  rename(Brent = Brent_Price_USD) |>
  mutate(
    Date = as.Date(Date),
    Brent = as.numeric(Brent)
  ) |>
  arrange(Date)
```

```r
cat("Brent date range:", min(brent_df$Date), "to", max(brent_df$Date), "\n")
```

```
Brent date range: 8404 to 20535
```

```r
# Helper to create weekly ts objects
make_weekly_ts <- function(x, start_date) {
  ts(
    x,
    start = c(
      as.numeric(format(start_date, "%Y")),
      as.numeric(format(start_date, "%U")) + 1
    ),
    frequency = 52
  )
}

gas_start_date <- min(gas_df$Date)

# Univariate gasoline series
us_gas_ts <- make_weekly_ts(gas_df$US_All_Grades, gas_start_date)
east_ts <- make_weekly_ts(gas_df$East_Coast, gas_start_date)
midwest_ts <- make_weekly_ts(gas_df$Midwest, gas_start_date)
gulf_ts <- make_weekly_ts(gas_df$Gulf_Coast, gas_start_date)
rocky_ts <- make_weekly_ts(gas_df$Rocky_Mountain, gas_start_date)
west_ts <- make_weekly_ts(gas_df$West_Coast, gas_start_date)

# Multivariate gasoline series for regional VAR analysis
region_var_ts <- gas_df |>
  dplyr::select(East_Coast, Midwest, Gulf_Coast, Rocky_Mountain, West_Coast) |>
  as.matrix() |>
  ts(
    start = c(
      as.numeric(format(gas_start_date, "%Y")),
      as.numeric(format(gas_start_date, "%U")) + 1
    ),
    frequency = 52
  )

colnames(region_var_ts) <- c(
  "East_Coast", "Midwest", "Gulf_Coast", "Rocky_Mountain", "West_Coast"
)
```

```r
# Align gasoline and Brent crude on common weekly dates
gas_brent_df <- gas_df |>
  inner_join(brent_df, by = "Date") |>
  drop_na()

cat(
  "Gasoline-Brent overlap:",
  format(min(gas_brent_df$Date), "%Y-%m-%d"), "to",
  format(max(gas_brent_df$Date), "%Y-%m-%d"), "\n"
)
```

```
Gasoline-Brent overlap: 1993-04-05 to 2026-03-23
```

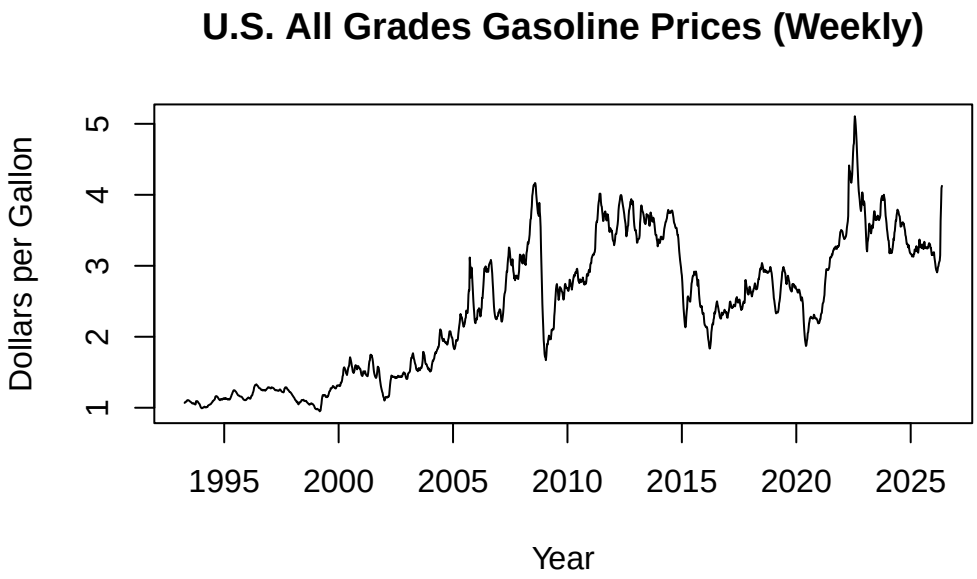
```r
cat("Overlap observations:", nrow(gas_brent_df), "\n")
```

```
Overlap observations: 1721
```

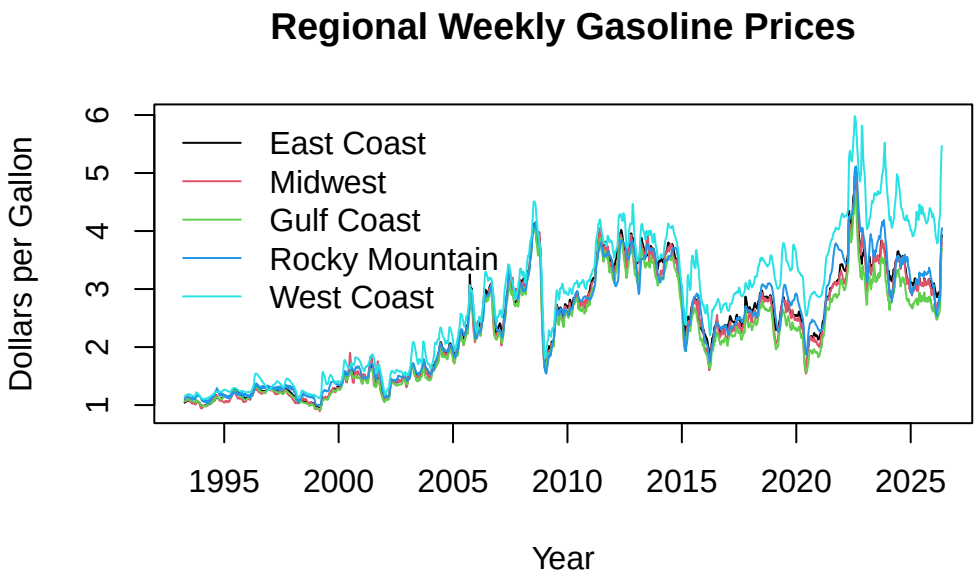
```r
gas_brent_start_date <- min(gas_brent_df$Date)

gas_ts_brent_overlap <- make_weekly_ts(gas_brent_df$US_All_Grades, gas_brent_start_date)
brent_ts <- make_weekly_ts(gas_brent_df$Brent, gas_brent_start_date)
```

2. Exploratory Data Analysis

```r
# National gasoline prices
plot(
  us_gas_ts,
  main = "U.S. All Grades Gasoline Prices (Weekly)",
  xlab = "Year",
  ylab = "Dollars per Gallon"
)
```

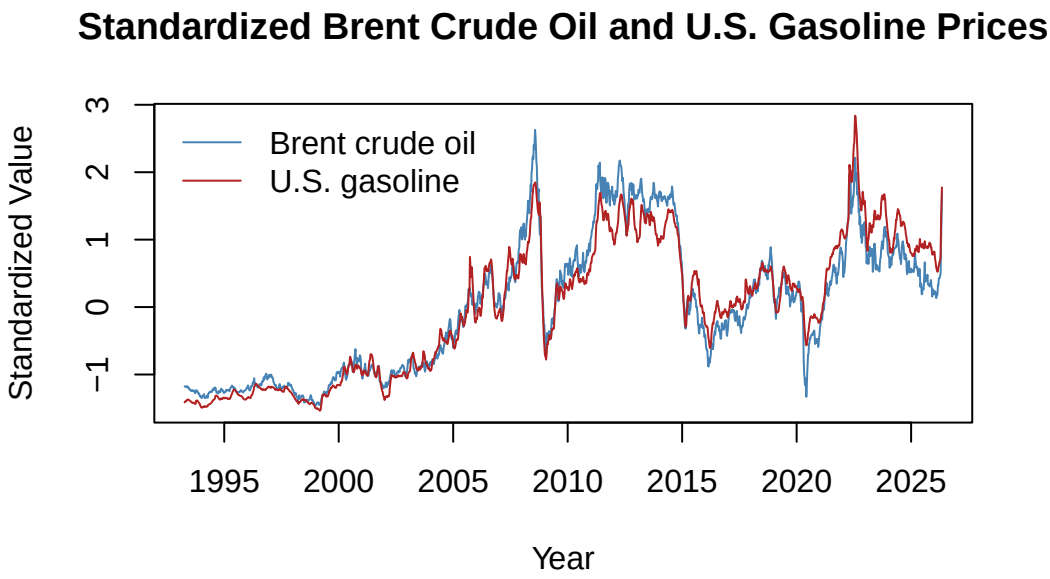



```r
# Regional gasoline prices
ts.plot(
  east_ts, midwest_ts, gulf_ts, rocky_ts, west_ts,
  col = 1:5, lty = 1,
  main = "Regional Weekly Gasoline Prices",
  xlab = "Year",
  ylab = "Dollars per Gallon"
)
legend(
  "topleft",
  legend = c("East Coast", "Midwest", "Gulf Coast", "Rocky Mountain", "West Coast"),
  col = 1:5, lty = 1, bty = "n"
)
```




```r
# Standardize gasoline and Brent for visual comparison
gas_scaled_ts <- ts(
  as.numeric(scale(as.numeric(gas_ts_brent_overlap))),
  start = start(gas_ts_brent_overlap),
  frequency = frequency(gas_ts_brent_overlap)
)

brent_scaled_ts <- ts(
  as.numeric(scale(as.numeric(brent_ts))),
  start = start(brent_ts),
  frequency = frequency(brent_ts)
)

ts.plot(
  brent_scaled_ts, gas_scaled_ts,
  col = c("steelblue", "firebrick"), lty = 1,
  main = "Standardized Brent Crude Oil and U.S. Gasoline Prices",
  xlab = "Year",
  ylab = "Standardized Value"
)
legend(
  "topleft",
  legend = c("Brent crude oil", "U.S. gasoline"),
  col = c("steelblue", "firebrick"),
  lty = 1,
  bty = "n"
```

```
)
```

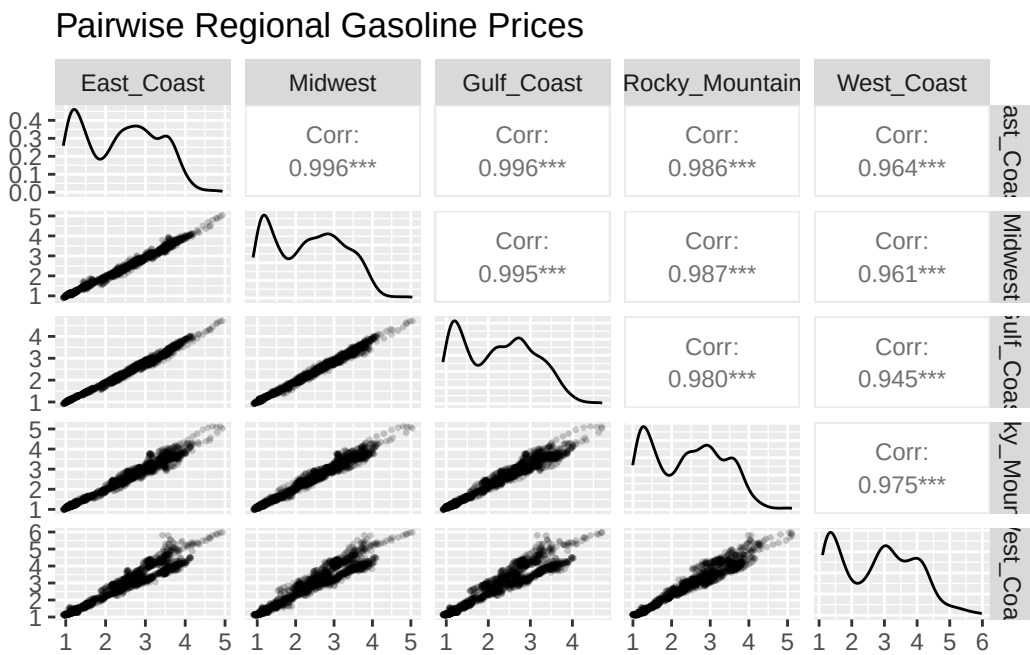


The national gasoline series shows long swings and persistent level changes rather than fluctuations around a fixed mean. That pattern suggests the level series is nonstationary and should be transformed before lag-based modeling.

The five regional series move together closely during major price episodes, but they are not identical. The West Coast is usually the highest-priced region and appears more volatile, while the Gulf Coast and Midwest stay closer to the national pattern and tend to lie at lower price levels.

The standardized gasoline and Brent series also move closely together over time. This supports the idea that crude oil is an important input into retail gasoline pricing, while still leaving room for pass-through delays and region-specific effects.

```
# Pairwise contemporaneous relationships across regions
gas_df |>
  dplyr::select(East_Coast, Midwest, Gulf_Coast, Rocky_Mountain, West_Coast) |>
  ggpairs(
    title = "Pairwise Regional Gasoline Prices",
    lower = list(continuous = wrap("points", alpha = 0.2, size = 0.5)),
    upper = list(continuous = wrap("cor", size = 3))
  )
```



The pairwise scatterplots show strong positive same-week relationships across all five regions. The reported correlations are uniformly high, ranging from 0.945 to 0.996, so the regions clearly are not evolving independently.

Several pairs, especially East Coast and Midwest, fall very tightly along a line. Pairs involving the West Coast are slightly more dispersed, which is consistent with the greater regional variation seen in the time-series plot.

These plots only describe contemporaneous association. They motivate a multivariate time-series approach, but they do not by themselves tell us which region moves first.

2.5. AR Model for U.S. Gasoline

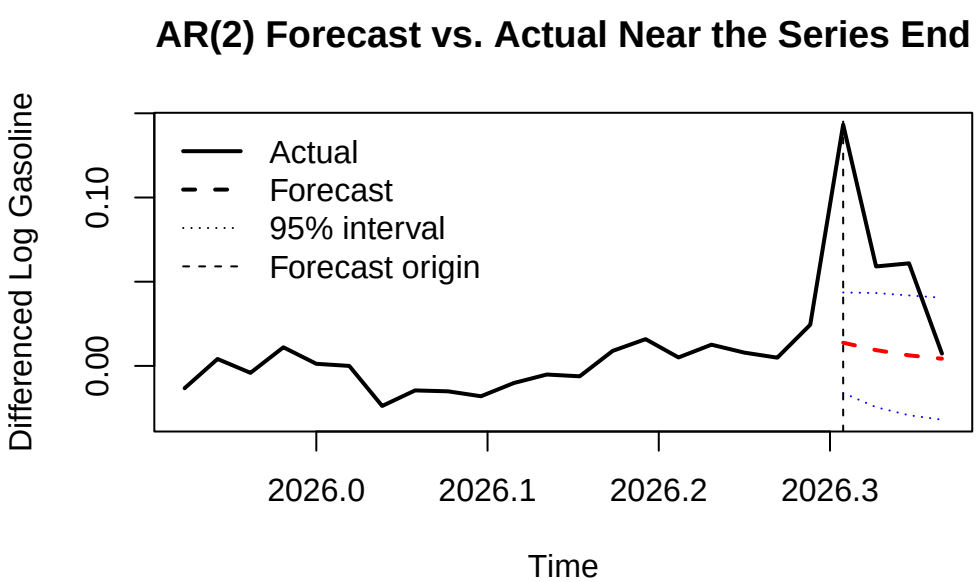
```
# Difference log gasoline prices to work with an approximately stationary series
d_us_gas <- diff(log(us_gas_ts))

# Hold out the last 4 weeks for a simple forecast check
forecast_horizon <- 4
n_gas <- length(d_us_gas)

train_d_us_gas <- window(d_us_gas, end = time(d_us_gas)[n_gas - forecast_horizon])
test_d_us_gas <- window(d_us_gas, start = time(d_us_gas)[n_gas - forecast_horizon + 1])
```

```r
# Fit AR(2) model
fit_ar2_gas <- arima(train_d_us_gas, order = c(2, 0, 0))
summary(fit_ar2_gas)
```

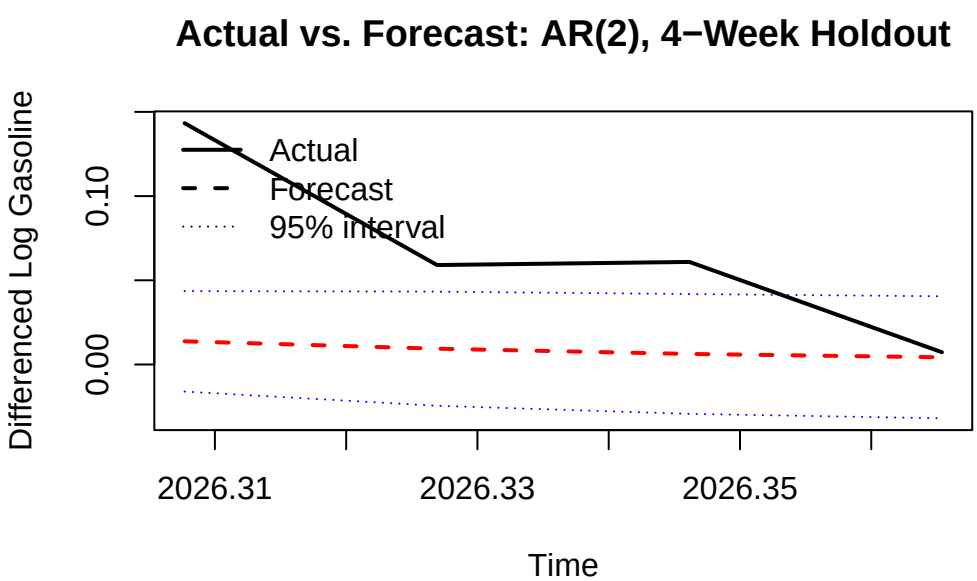
```
Call:
arima(x = train_d_us_gas, order = c(2, 0, 0))

Coefficients:
         ar1     ar2  intercept
      0.5395  0.0696      6e-04
s.e.  0.0241  0.0241      9e-04

sigma^2 estimated as 0.0002316:  log likelihood = 4749.41,  aic = -9490.82

Training set error measures:
                       ME      RMSE        MAE MPE MAPE      MASE         ACF1
Training set 6.608431e-06 0.0152196 0.01051354 NaN  Inf 0.9035163 -0.004852529
```

```r
# Forecast the held-out period
fc_ar2_gas <- forecast(fit_ar2_gas, h = forecast_horizon, level = 95)

fc_mean_ts <- ts(
  as.numeric(fc_ar2_gas$mean),
  start = start(test_d_us_gas),
  frequency = frequency(test_d_us_gas)
)

fc_lower_ts <- ts(
  as.numeric(fc_ar2_gas$lower[, 1]),
  start = start(test_d_us_gas),
  frequency = frequency(test_d_us_gas)
)

fc_upper_ts <- ts(
  as.numeric(fc_ar2_gas$upper[, 1]),
  start = start(test_d_us_gas),
  frequency = frequency(test_d_us_gas)
)

# Show recent history leading into the forecast origin
```

```r
weeks_of_history <- 20
tail_start_time <- time(d_us_gas)[n_gas - forecast_horizon - weeks_of_history + 1]
actual_tail_ts <- window(d_us_gas, start = tail_start_time)

plot(
  actual_tail_ts,
  type = "l",
  lwd = 2,
  main = "AR(2) Forecast vs. Actual Near the Series End",
  xlab = "Time",
  ylab = "Differenced Log Gasoline",
  ylim = range(c(actual_tail_ts, fc_mean_ts, fc_lower_ts, fc_upper_ts))
)
lines(fc_mean_ts, lwd = 2, lty = 2, col = "red")
lines(fc_lower_ts, lty = 3, col = "blue")
lines(fc_upper_ts, lty = 3, col = "blue")
abline(v = time(test_d_us_gas)[1], lty = 2)
legend(
  "topleft",
  legend = c("Actual", "Forecast", "95% interval", "Forecast origin"),
  lty = c(1, 2, 3, 2),
  lwd = c(2, 2, 1, 1),
  bty = "n"
)
```



```r
# Plot held-out actual values against forecasts
plot(
  test_d_us_gas,
  type = "l",
  lwd = 2,
  main = "Actual vs. Forecast: AR(2), 4-Week Holdout",
  xlab = "Time",
  ylab = "Differenced Log Gasoline",
  ylim = range(c(test_d_us_gas, fc_mean_ts, fc_lower_ts, fc_upper_ts))
)
lines(fc_mean_ts, lwd = 2, lty = 2, col = "red")
lines(fc_lower_ts, lty = 3, col = "blue")
lines(fc_upper_ts, lty = 3, col = "blue")
legend(
  "topleft",
  legend = c("Actual", "Forecast", "95% interval"),
  lty = c(1, 2, 3),
  lwd = c(2, 2, 1),
  bty = "n"
)
```



The AR(2) fit suggests that gasoline price changes are driven mostly by very recent history. The lag 1 coefficient is 0.5395, while the lag 2 coefficient is only 0.0696, so most of the predictable signal comes from the previous week rather than two weeks back.

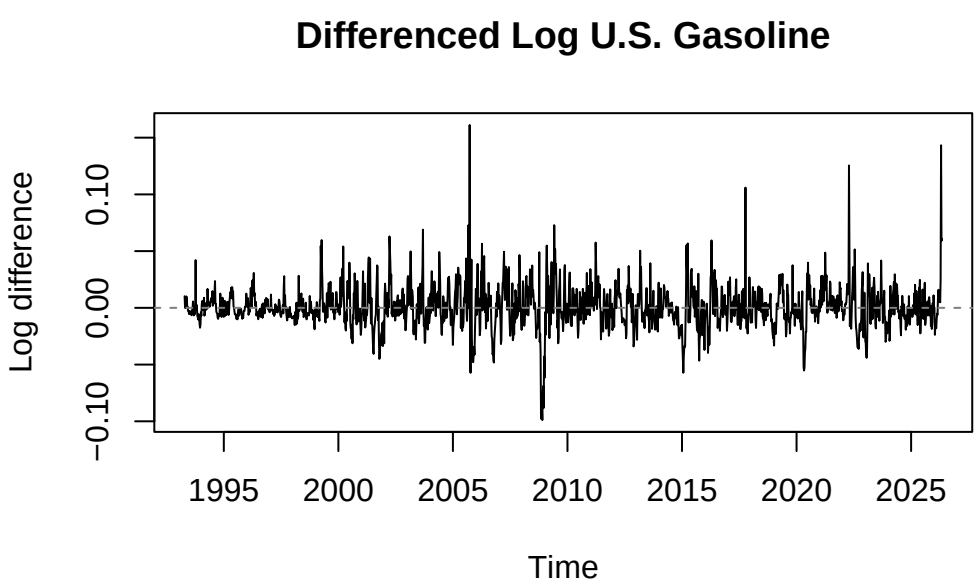
The fitted mean is 0.0006, which is close to zero after differencing and logging. The residual lag 1 autocorrelation is -0.0049, so there is little obvious linear dependence left in the residuals.

The forecast plots show the main limitation of this univariate model. The 4-week forecasts quickly revert toward zero and miss the sharp movement at the end of the sample. Even though the observed values remain inside the forecast interval, the point forecasts are too smooth to track the sudden change. This suggests that a model based only on past gasoline prices is not enough when market conditions shift quickly.

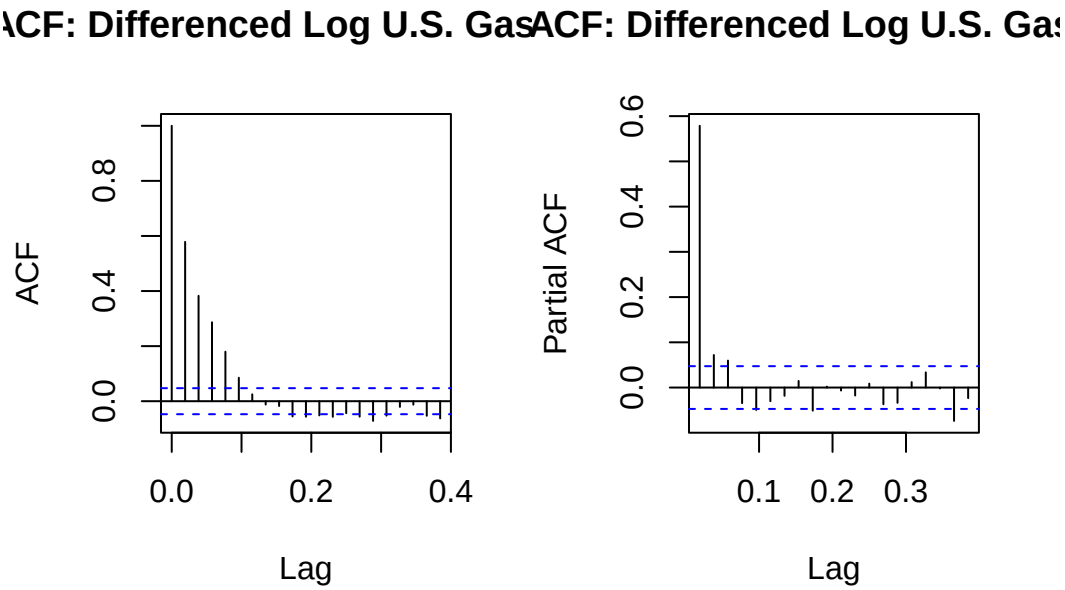
3. Brent Crude Oil vs. U.S. Gasoline

(i) Convert to approximately stationary series

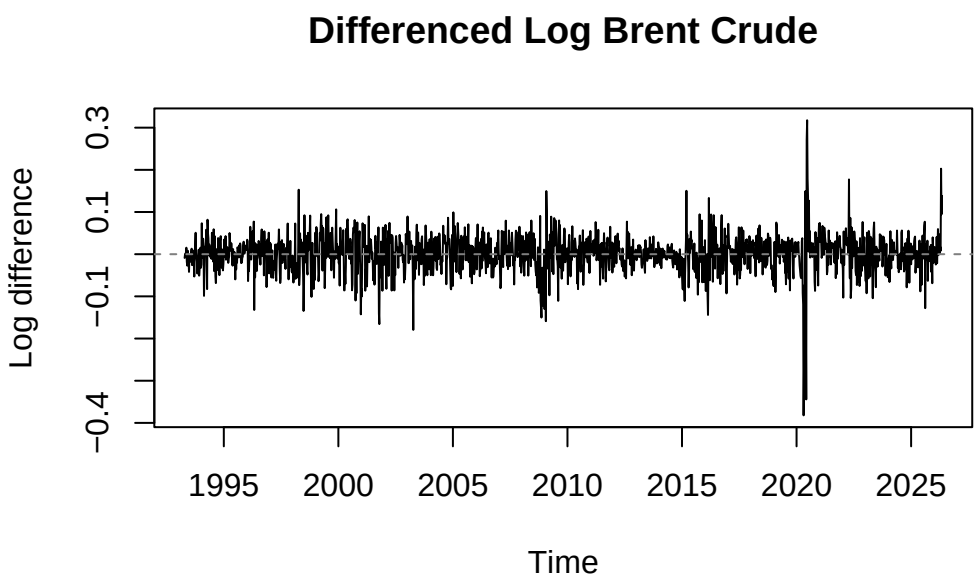
```
# First differences of log prices
d_us_gas_overlap <- diff(log(gas_ts_brent_overlap))
d_brent <- diff(log(brent_ts))

# Calendar dates corresponding to the differenced series
gas_brent_diff_dates <- gas_brent_df$Date[-1]

# Differenced log U.S. gasoline
plot(
  d_us_gas_overlap,
  main = "Differenced Log U.S. Gasoline",
  ylab = "Log difference",
  xlab = "Time"
)
abline(h = 0, lty = 2, col = "gray50")
```



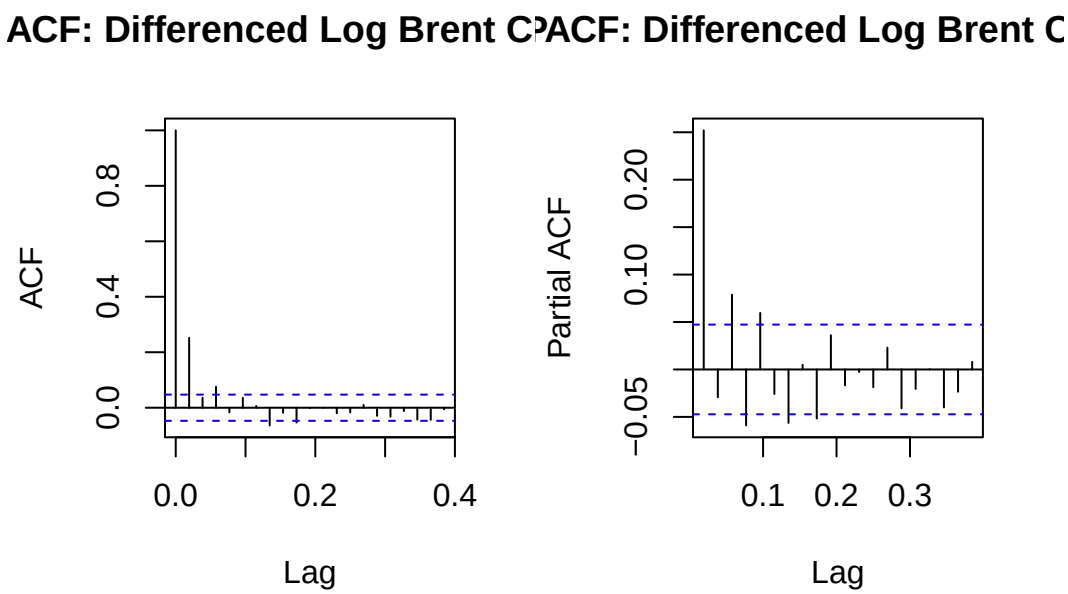
```r
par(mfrow = c(1, 2))
acf(d_us_gas_overlap, lag.max = 20,
  main = "ACF: Differenced Log U.S. Gasoline")

pacf(d_us_gas_overlap, lag.max = 20, main = "PACF: Differenced Log U.S. Gasoline")
```




```r
par(mfrow = c(1, 1))

# Differenced log Brent crude
plot(
  d_brent,
  main = "Differenced Log Brent Crude",
  ylab = "Log difference",
  xlab = "Time"
)
abline(h = 0, lty = 2, col = "gray50")
```




```
par(mfrow = c(1, 2))
acf(d_brent, lag.max = 20, main = "ACF: Differenced Log Brent Crude")
pacf(d_brent, lag.max = 20, main = "PACF: Differenced Log Brent Crude")
```



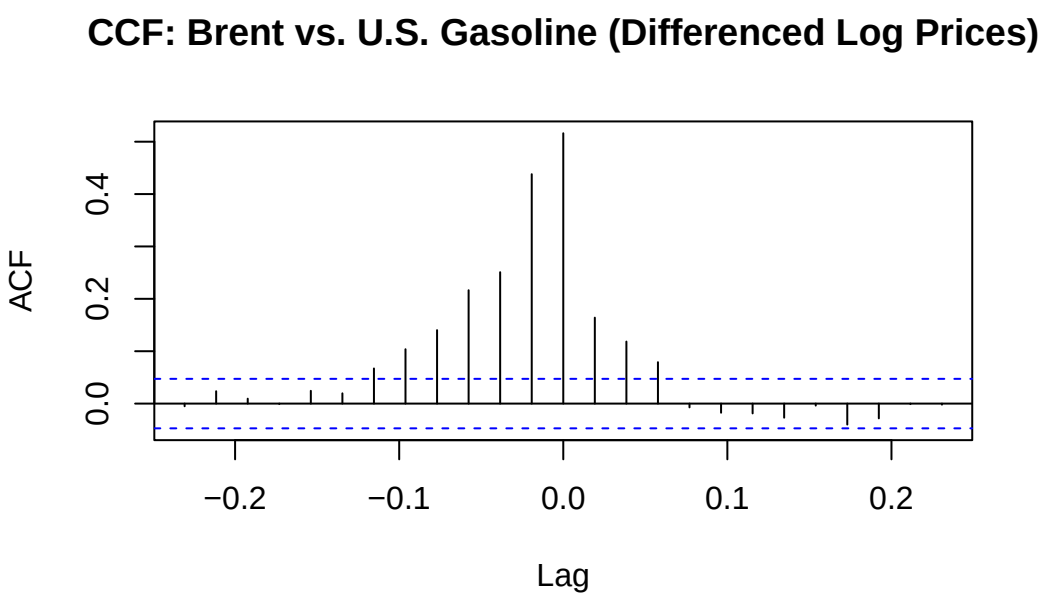

```
par(mfrow = c(1, 1))
```

After differencing the log prices, both series fluctuate around a more stable mean and the autocorrelations decay much more quickly than in levels. The PACFs are dominated by the

first few lags, which is consistent with using low-order dynamic models for the transformed series.

(ii) CCF and lagged regression

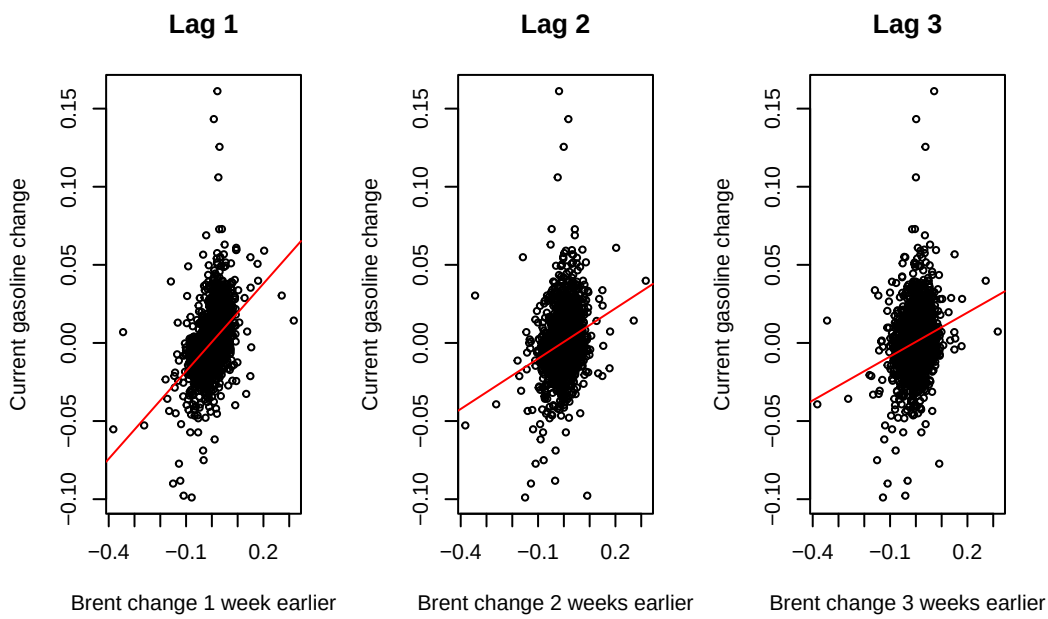
```
# Negative lag means Brent leads gasoline
ccf(
  d_brent, d_us_gas_overlap,
  lag.max = 12,
  main = "CCF: Brent vs. U.S. Gasoline (Differenced Log Prices)"
)
```



The cross-correlation function peaks at lag 0, which means gasoline and Brent often move strongly within the same week. There are also positive correlations at negative lags, especially around -1 to -3, which is consistent with Brent changes showing up in gasoline prices over the following weeks.

```
# Build a lagged regression data set
oil_gas_lag_df <- ts.intersect(
  gas = d_us_gas_overlap,
  brent_lag_1 = stats::lag(d_brent, -1),
  brent_lag_2 = stats::lag(d_brent, -2),
  brent_lag_3 = stats::lag(d_brent, -3),
  dframe = TRUE
```

```r
)

par(mfrow = c(1, 3))

plot(
  oil_gas_lag_df$brent_lag_1, oil_gas_lag_df$gas,
  xlab = "Brent change 1 week earlier",
  ylab = "Current gasoline change",
  main = "Lag 1"
)
abline(lm(gas ~ brent_lag_1, data = oil_gas_lag_df), col = "red")

plot(
  oil_gas_lag_df$brent_lag_2, oil_gas_lag_df$gas,
  xlab = "Brent change 2 weeks earlier",
  ylab = "Current gasoline change",
  main = "Lag 2"
)
abline(lm(gas ~ brent_lag_2, data = oil_gas_lag_df), col = "red")

plot(
  oil_gas_lag_df$brent_lag_3, oil_gas_lag_df$gas,
  xlab = "Brent change 3 weeks earlier",
  ylab = "Current gasoline change",
  main = "Lag 3"
)
abline(lm(gas ~ brent_lag_3, data = oil_gas_lag_df), col = "red")
```




```r
par(mfrow = c(1, 1))
```

```r
# Regression using Brent lags 1 to 3
fit_oil_lag_lm <- lm(
  gas ~ brent_lag_1 + brent_lag_2 + brent_lag_3,
  data = oil_gas_lag_df
)

summary(fit_oil_lag_lm)
```

```
Call:
lm(formula = gas ~ brent_lag_1 + brent_lag_2 + brent_lag_3, data = oil_gas_lag_df)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.080184 -0.008914 -0.001070  0.007719  0.152296 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.0005041  0.0004021   1.254     0.21    
brent_lag_1 0.1735182  0.0092325  18.794  < 2e-16 ***
brent_lag_2 0.0449195  0.0095342   4.711 2.66e-06 ***
brent_lag_3 0.0776113  0.0092899   8.354  < 2e-16 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01665 on 1713 degrees of freedom
Multiple R-squared:  0.2452,    Adjusted R-squared:  0.2439
F-statistic: 185.5 on 3 and 1713 DF,  p-value: < 2.2e-16
```

The lagged scatterplots support the CCF results. The association is strongest at a one-week lag, then becomes weaker at two and three weeks. In practical terms, this suggests that gasoline prices do not adjust all at once; some of the crude oil effect appears to pass through with a short delay.

(iii) VAR(2) model

```r
# Bivariate stationary series for VAR analysis
oil_var_ts <- cbind(
  Gasoline = d_us_gas_overlap,
  Brent = d_brent
)

# Candidate lag lengths
VARselect(oil_var_ts, lag.max = 8, type = "none")
```

```
$selection
AIC(n)  HQ(n)  SC(n) FPE(n)
     5      3      2      5

$criteria
                   1             2             3             4             5
AIC(n) -1.491150e+01 -1.492793e+01 -1.493507e+01 -1.493885e+01 -1.494229e+01
HQ(n)  -1.490679e+01 -1.491851e+01 -1.492094e+01 -1.492001e+01 -1.491875e+01
SC(n)  -1.489878e+01 -1.490248e+01 -1.489690e+01 -1.488796e+01 -1.487867e+01
FPE(n)  3.342081e-07  3.287641e-07  3.264252e-07  3.251927e-07  3.240755e-07
                   6             7             8
AIC(n) -1.493893e+01 -1.493895e+01 -1.493744e+01
HQ(n)  -1.491068e+01 -1.490599e+01 -1.489977e+01
SC(n)  -1.486260e+01 -1.484989e+01 -1.483566e+01
FPE(n)  3.251649e-07  3.251606e-07  3.256512e-07
```

```r
# Fit the working VAR(2) model
fit_var_oil <- VAR(oil_var_ts, p = 2, type = "none")
summary(fit_var_oil)
```

```
VAR Estimation Results:
=========================
Endogenous variables: Gasoline, Brent
Deterministic variables: none
Sample size: 1718
Log Likelihood: 7960.895
Roots of the characteristic polynomial:
0.6704 0.2709 0.2709 0.1272
Call:
VAR(y = oil_var_ts, p = 2, type = "none")


Estimation results for equation Gasoline:
=========================================
Gasoline = Gasoline.l1 + Brent.l1 + Gasoline.l2 + Brent.l2

             Estimate Std. Error t value Pr(>|t|)
Gasoline.l1  0.411722   0.027724  14.851  < 2e-16 ***
Brent.l1     0.093528   0.009721   9.622  < 2e-16 ***
Gasoline.l2  0.141533   0.025792   5.488 4.69e-08 ***
Brent.l2    -0.023122   0.009733  -2.375   0.0176 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


Residual standard error: 0.01515 on 1714 degrees of freedom
Multiple R-Squared: 0.3757, Adjusted R-squared: 0.3742
F-statistic: 257.8 on 4 and 1714 DF,  p-value: < 2.2e-16


Estimation results for equation Brent:
======================================
Brent = Gasoline.l1 + Brent.l1 + Gasoline.l2 + Brent.l2

             Estimate Std. Error t value Pr(>|t|)
Gasoline.l1  0.006147   0.079397   0.077 0.938301
Brent.l1     0.255444   0.027838   9.176  < 2e-16 ***
```

```
Gasoline.l2  0.294836   0.073864   3.992 6.84e-05 ***
Brent.l2    -0.092385   0.027875  -3.314 0.000938 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


Residual standard error: 0.0434 on 1714 degrees of freedom
Multiple R-Squared: 0.07644,    Adjusted R-squared: 0.07429
F-statistic: 35.47 on 4 and 1714 DF,  p-value: < 2.2e-16



Covariance matrix of residuals:
          Gasoline     Brent
Gasoline 0.0002296 0.0003275
Brent    0.0003275 0.0018832

Correlation matrix of residuals:
         Gasoline  Brent
Gasoline   1.0000 0.4981
Brent      0.4981 1.0000
```

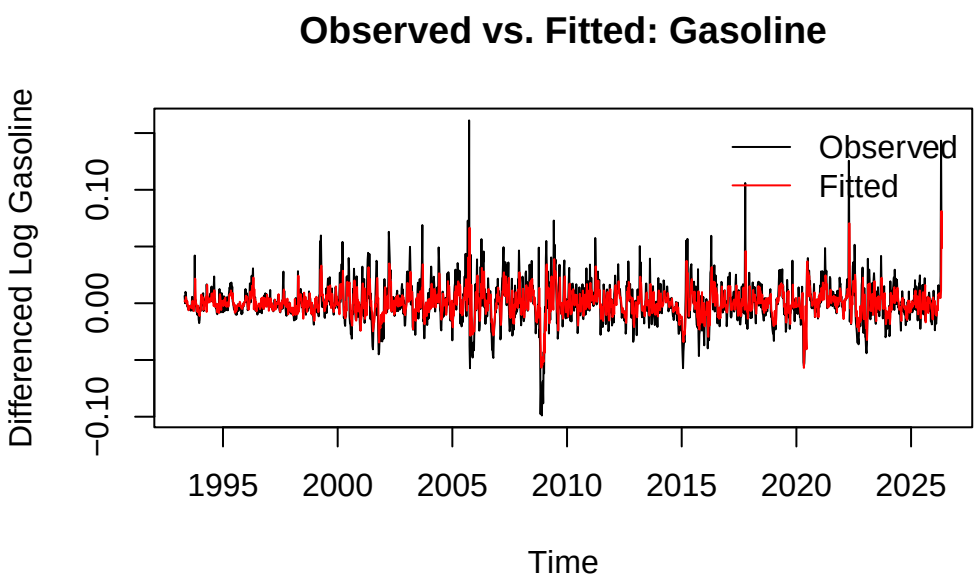
We use a VAR(2) as a simple low-order model for the transformed gasoline and Brent series. This specification is consistent with the short-memory pattern in the ACF, PACF, and CCF plots and gives a direct way to estimate whether each series helps predict the other.

(iv) Diagnostics

```r
# Fitted values and residuals
fitted_oil <- fitted(fit_var_oil)
resid_oil  <- resid(fit_var_oil)
```

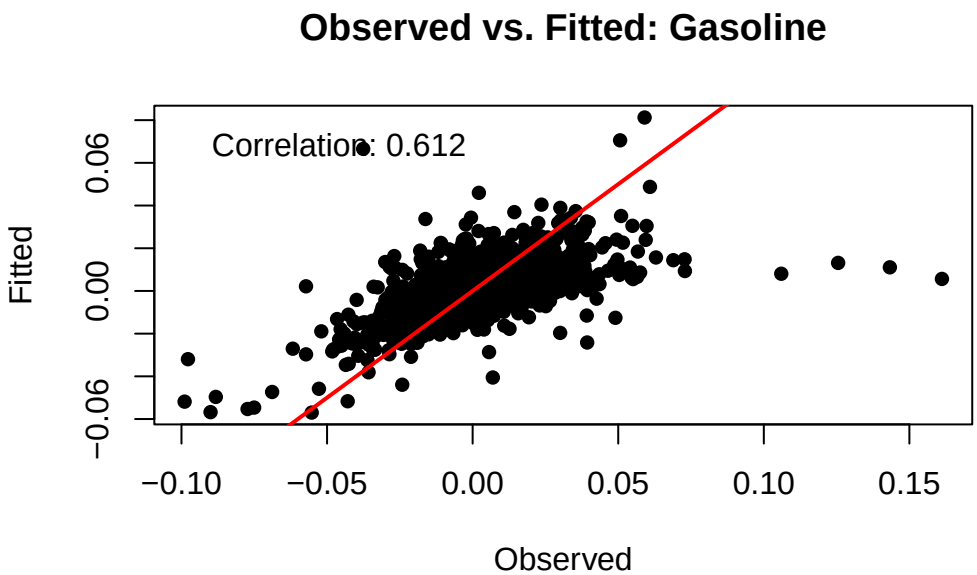
```r
# Observed and fitted gasoline changes
obs_gas_oil <- window(oil_var_ts[, "Gasoline"], start = time(oil_var_ts)[3])
fit_gas_oil <- ts(
  fitted_oil[, "Gasoline"],
  start = start(obs_gas_oil),
  frequency = frequency(obs_gas_oil)
)
```

```r
plot(
  obs_gas_oil, type = "l", col = "black",
  main = "Observed vs. Fitted: Gasoline",
  ylab = "Differenced Log Gasoline",
  xlab = "Time"
)
lines(fit_gas_oil, col = "red")
legend(
  "topright",
  legend = c("Observed", "Fitted"),
  col = c("black", "red"),
  lwd = 1,
  bty = "n"
)
```




```r
plot(
  obs_gas_oil, fit_gas_oil,
  pch = 16,
  xlab = "Observed",
  ylab = "Fitted",
  main = "Observed vs. Fitted: Gasoline"
)
legend(
  "topleft",
  legend = paste("Correlation:", round(cor(obs_gas_oil, fit_gas_oil), 3)),
```

```r
  bty = "n"
)
abline(0, 1, col = "red", lwd = 2)
```




```r
# Largest absolute residuals for gasoline
resid_gas_oil <- resid_oil[, "Gasoline"]
oil_resid_time <- time(oil_var_ts)[-c(1, 2)]

top_resid_gas_idx <- order(abs(resid_gas_oil), decreasing = TRUE)[1:10]
top_resid_gas_dates <- oil_resid_time[top_resid_gas_idx] |>
  date_decimal() |>
  as.Date()

tibble(
  Date = top_resid_gas_dates,
  Gasoline_Residual = resid_gas_oil[top_resid_gas_idx]
)
```

```
# A tibble: 10 x 2
   Date       Gasoline_Residual
   <date>                 <dbl>
 1 2005-09-24             0.156
 2 2026-04-23             0.132
 3 2022-04-16             0.112
 4 2005-10-01            -0.104
```

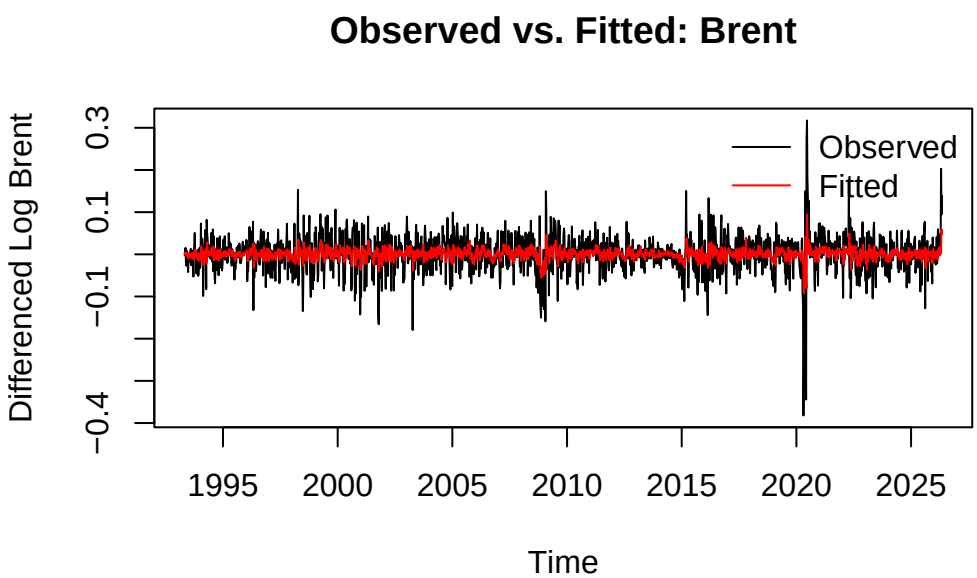
```
 5 2017-10-08            0.0979
 6 2008-11-05           -0.0658
 7 2009-06-04            0.0636
 8 2009-01-29            0.0635
 9 2008-10-08            0.0616
10 2005-10-08           -0.0594
```

```r
# Largest absolute residuals for Brent
resid_brent_oil <- resid_oil[, "Brent"]

top_resid_brent_idx <- order(abs(resid_brent_oil), decreasing = TRUE)[1:10]
top_resid_brent_dates <- oil_resid_time[top_resid_brent_idx] |>
  date_decimal() |>
  as.Date()

tibble(
  Date = top_resid_brent_dates,
  Brent_Residual = resid_brent_oil[top_resid_brent_idx]
)
```

```
# A tibble: 10 x 2
   Date       Brent_Residual
   <date>              <dbl>
 1 2020-04-22         -0.355
 2 2020-06-10          0.352
 3 2020-06-03         -0.283
 4 2020-06-17          0.223
 5 2020-05-20          0.202
 6 2026-04-23          0.201
 7 2020-05-27         -0.188
 8 2009-01-29          0.188
 9 2020-04-29         -0.170
10 2003-04-09         -0.170
```

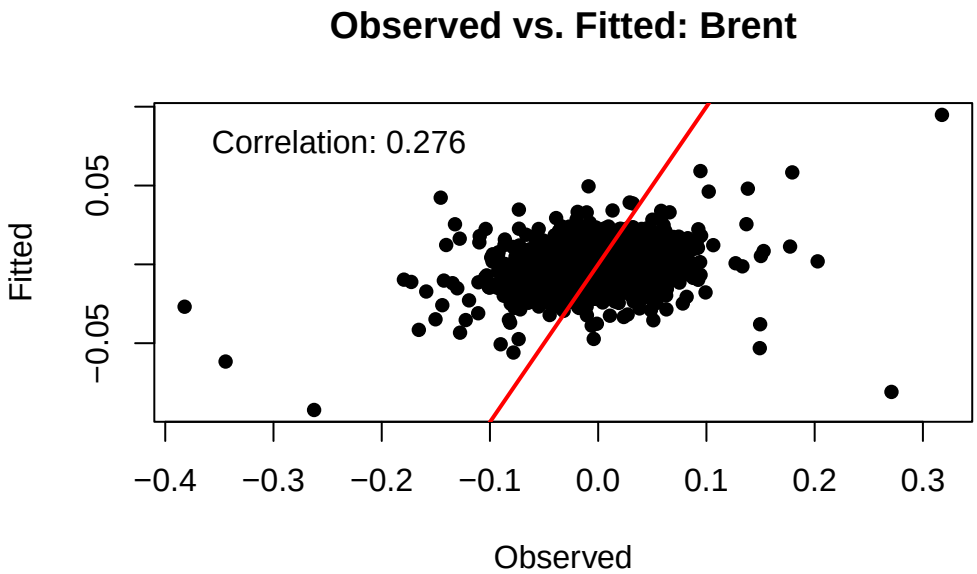
```r
# Observed and fitted Brent changes
obs_brent_oil <- window(oil_var_ts[, "Brent"], start = time(oil_var_ts)[3])
fit_brent_oil <- ts(
  fitted_oil[, "Brent"],
  start = start(obs_brent_oil),
  frequency = frequency(obs_brent_oil)
)
```

```r
plot(
  obs_brent_oil, type = "l", col = "black",
  main = "Observed vs. Fitted: Brent",
  ylab = "Differenced Log Brent",
  xlab = "Time"
)
lines(fit_brent_oil, col = "red")
legend(
  "topright",
  legend = c("Observed", "Fitted"),
  col = c("black", "red"),
  lwd = 1,
  bty = "n"
)
```

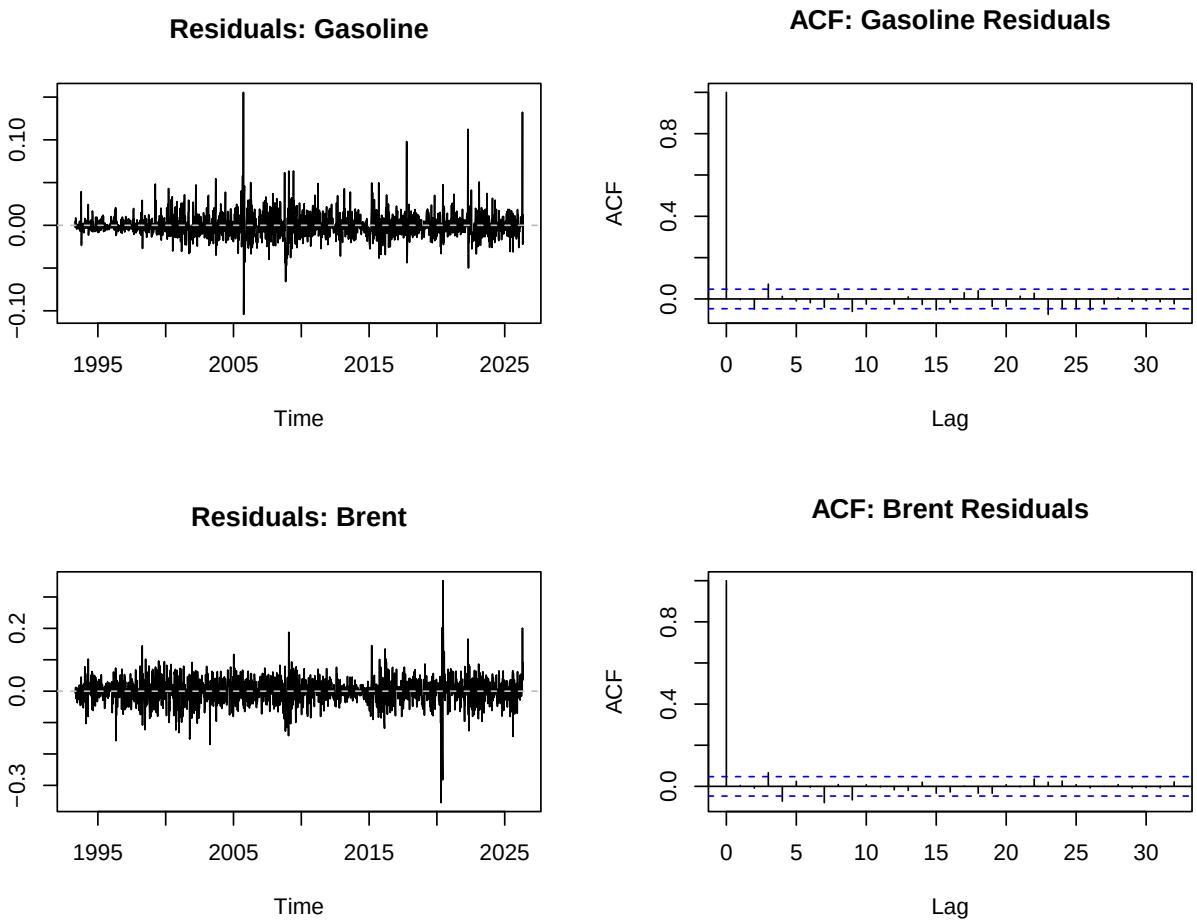



```r
plot(
  obs_brent_oil, fit_brent_oil,
  pch = 16,
  xlab = "Observed",
  ylab = "Fitted",
  main = "Observed vs. Fitted: Brent"
)
legend(
  "topleft",
  legend = paste("Correlation:", round(cor(obs_brent_oil, fit_brent_oil), 3)),
```

```
  bty = "n"
)
abline(0, 1, col = "red", lwd = 2)
```

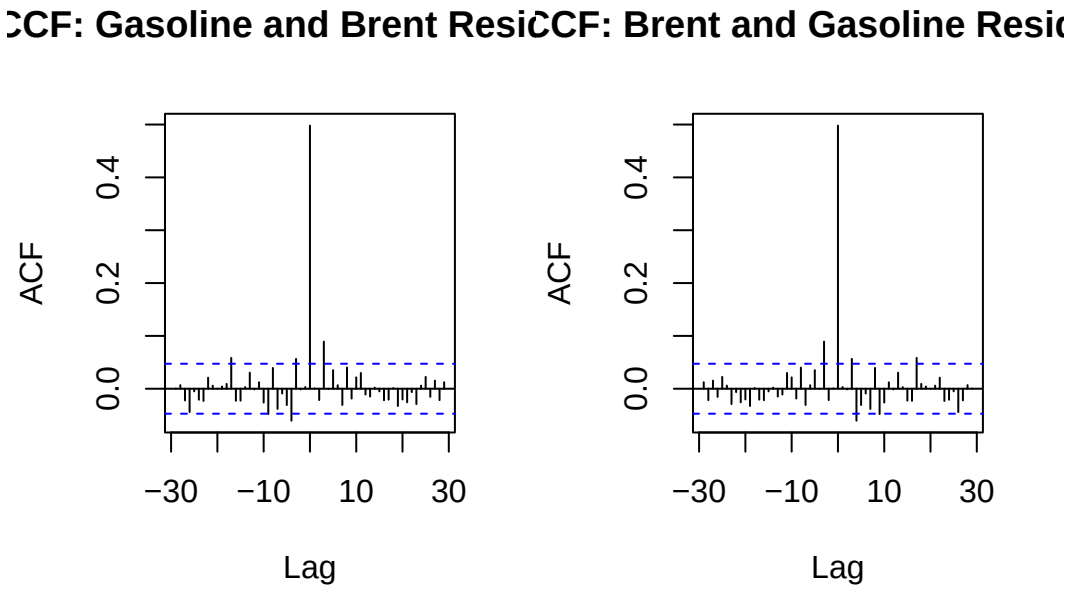



```
# Residual series
resid_gas_ts <- ts(
  resid_oil[, "Gasoline"],
  start = time(oil_var_ts)[3],
  frequency = frequency(oil_var_ts)
)

resid_brent_ts <- ts(
  resid_oil[, "Brent"],
  start = time(oil_var_ts)[3],
  frequency = frequency(oil_var_ts)
)

par(mfrow = c(2, 2))

plot(resid_gas_ts, type = "l", main = "Residuals: Gasoline", ylab = "", xlab = "Time")
abline(h = 0, lty = 2, col = "gray")
acf(resid_oil[, "Gasoline"], main = "ACF: Gasoline Residuals")

plot(resid_brent_ts, type = "l", main = "Residuals: Brent", ylab = "", xlab = "Time")
abline(h = 0, lty = 2, col = "gray")
acf(resid_oil[, "Brent"], main = "ACF: Brent Residuals")
```



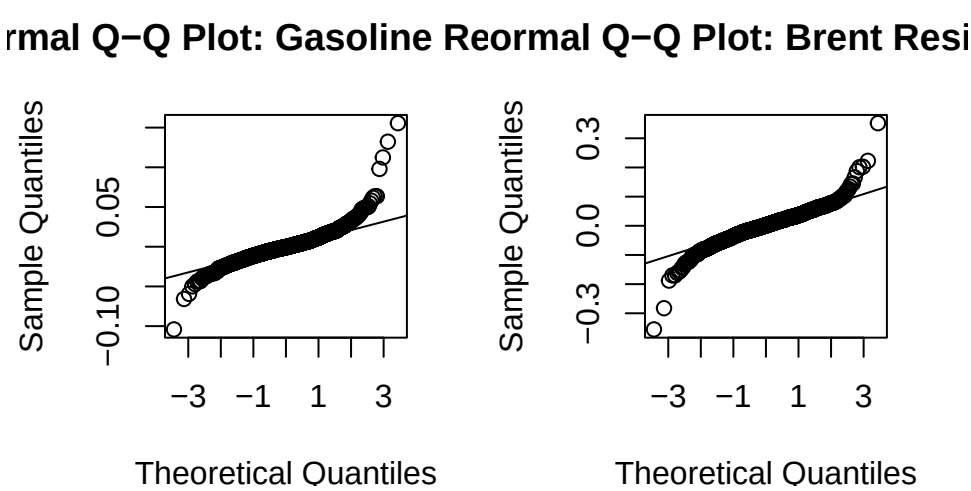

```r
par(mfrow = c(1, 1))
```

```r
# Residual cross-correlation
par(mfrow = c(1, 2))

ccf(resid_oil[, "Gasoline"], resid_oil[, "Brent"],
    main = "CCF: Gasoline and Brent Residuals")

ccf(resid_oil[, "Brent"], resid_oil[, "Gasoline"],
    main = "CCF: Brent and Gasoline Residuals")
```




```r
par(mfrow = c(1, 1))
```

```r
# Normal Q-Q plots
par(mfrow = c(1, 2))

qqnorm(resid_oil[, "Gasoline"], main = "Normal Q-Q Plot: Gasoline Residuals")
qqline(resid_oil[, "Gasoline"])

qqnorm(resid_oil[, "Brent"], main = "Normal Q-Q Plot: Brent Residuals")
qqline(resid_oil[, "Brent"])
```



```r
par(mfrow = c(1, 1))
```

Overall, the VAR(2) captures the broad short-run dependence reasonably well, especially for gasoline. The fitted-versus-observed plots show that the model tracks the center of the series, but it still misses the largest shocks, which is what we would expect from a low-order linear model fit to differenced prices.

The residual series fluctuate around zero and the residual ACFs show little remaining serial structure. The main departures appear in the tails of the Q-Q plots, especially for Brent, which suggests that unusual market weeks are still harder to explain than ordinary weeks.

(v) Granger causality

```r
causality(fit_var_oil, cause = "Brent")
```

```
$Granger

    Granger causality H0: Brent do not Granger-cause Gasoline

data:  VAR object fit_var_oil
F-Test = 47.151, df1 = 2, df2 = 3428, p-value < 2.2e-16


$Instant

    H0: No instantaneous causality between: Brent and Gasoline

data:  VAR object fit_var_oil
Chi-squared = 341.64, df = 1, p-value < 2.2e-16
```

```r
causality(fit_var_oil, cause = "Gasoline")
```

```
$Granger

    Granger causality H0: Gasoline do not Granger-cause Brent

data:  VAR object fit_var_oil
F-Test = 10.806, df1 = 2, df2 = 3428, p-value = 2.097e-05
```

```
$Instant

    H0: No instantaneous causality between: Gasoline and Brent

data:  VAR object fit_var_oil
Chi-squared = 341.64, df = 1, p-value < 2.2e-16
```

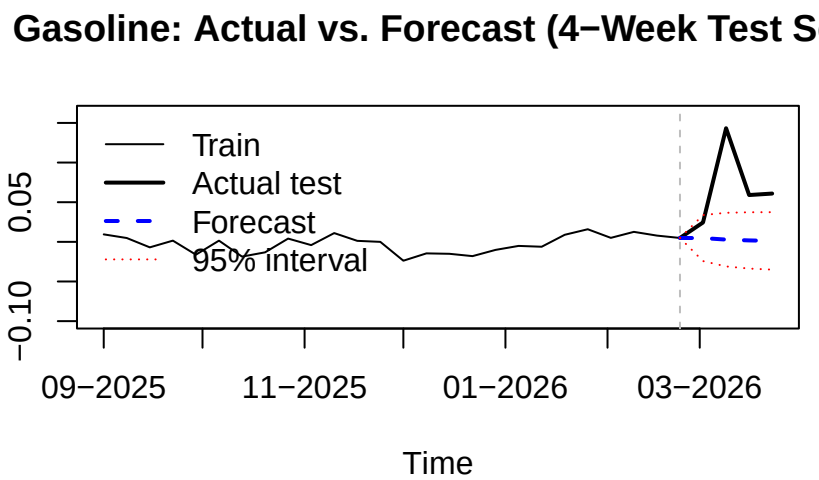
The Granger tests are best interpreted as predictive tests, not as proof of structural causation. In this context, bidirectional significance would mean that each series contains useful lagged information about the other, while a stronger Brent-to-gasoline result would still be more consistent with the economic story that crude oil price movements pass through into retail gasoline prices.

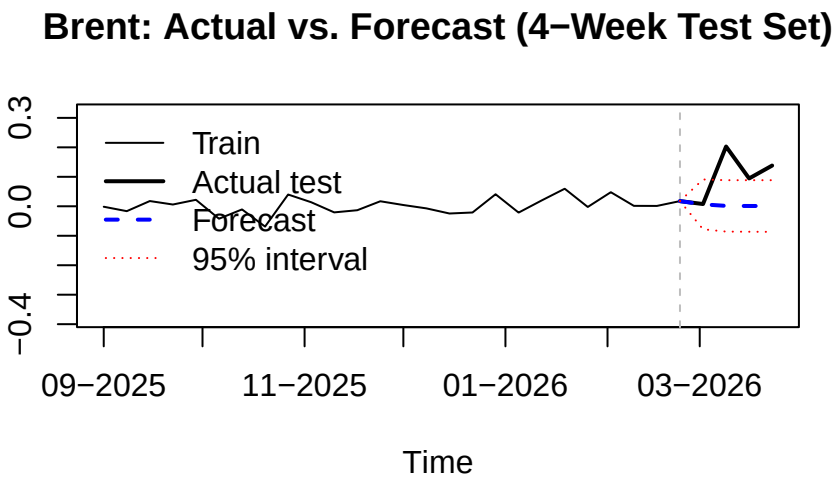
(vi) 4-week holdout forecast

```r
# Hold out the last 4 weeks
forecast_horizon <- 4

oil_train_ts <- oil_var_ts[1:(nrow(oil_var_ts) - forecast_horizon), ]
oil_test_ts  <- oil_var_ts[(nrow(oil_var_ts) - forecast_horizon + 1):nrow(oil_var_ts), ]

oil_train_dates <- gas_brent_diff_dates[1:(length(gas_brent_diff_dates) - forecast_horizon)]
oil_test_dates  <- gas_brent_diff_dates[(length(gas_brent_diff_dates) - forecast_horizon + 1

fit_var_oil_train <- VAR(oil_train_ts, p = 2, type = "none")
oil_forecast <- predict(fit_var_oil_train, n.ahead = forecast_horizon, ci = 0.95)

series_names <- c("Gasoline", "Brent")
series_titles <- c(
  "Gasoline: Actual vs. Forecast (4-Week Test Set)",
  "Brent: Actual vs. Forecast (4-Week Test Set)"
)

for (i in seq_along(series_names)) {
  series_name <- series_names[i]

  y_train <- oil_train_ts[, series_name]
  y_test  <- oil_test_ts[, series_name]
  fc_df   <- as.data.frame(oil_forecast$fcst[[series_name]])
```

```r
plot_start_idx <- max(1, length(oil_train_dates) - 26 + 1)
recent_dates <- oil_train_dates[plot_start_idx:length(oil_train_dates)]
recent_train_values <- y_train[plot_start_idx:length(y_train)]

plot(
  x = recent_dates,
  y = recent_train_values,
  type = "l",
  xlim = c(min(recent_dates), max(oil_test_dates)),
  ylim = range(c(y_train, y_test, fc_df$fcst, fc_df$lower, fc_df$upper), na.rm = TRUE),
  xlab = "Time",
  ylab = "",
  main = series_titles[i],
  xaxt = "n"
)

tick_pos <- seq(from = min(recent_dates), to = max(oil_test_dates), by = "1 month")
axis(1, at = tick_pos, labels = format(tick_pos, "%m-%Y"))

abline(v = tail(oil_train_dates, 1), lty = 2, col = "gray")

lines(
  x = c(tail(oil_train_dates, 1), oil_test_dates),
  y = c(tail(y_train, 1), y_test),
  col = "black",
  lwd = 2
)

lines(
  x = c(tail(oil_train_dates, 1), oil_test_dates),
  y = c(tail(y_train, 1), fc_df$fcst),
  col = "blue",
  lty = 2,
  lwd = 2
)

lines(
  x = c(tail(oil_train_dates, 1), oil_test_dates),
  y = c(tail(y_train, 1), fc_df$lower),
  col = "red",
  lty = 3
)
```

```r
  lines(
    x = c(tail(oil_train_dates, 1), oil_test_dates),
    y = c(tail(y_train, 1), fc_df$upper),
    col = "red",
    lty = 3
  )

  legend(
    "topleft",
    legend = c("Train", "Actual test", "Forecast", "95% interval"),
    col = c("black", "black", "blue", "red"),
    lty = c(1, 1, 2, 3),
    lwd = c(1, 2, 2, 1),
    bty = "n"
  )
}
```





The 4-week holdout forecast shows that the bivariate VAR is still too smooth to capture the sharp spike at the end of the sample. Relative to the univariate AR model, it uses more information, but it still behaves like a short-memory model for differenced prices and therefore pulls quickly back toward zero.

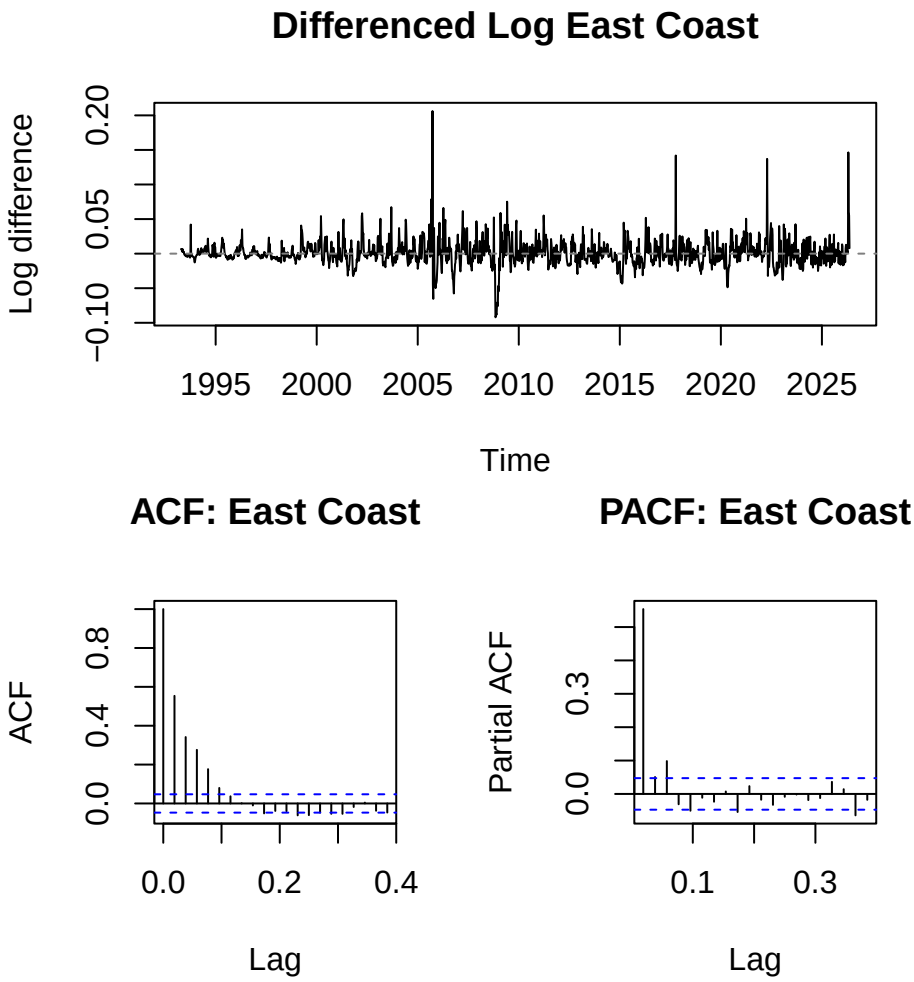
That result matters for interpretation. The oil-gasoline link is useful for understanding dependence and pass-through, but it does not automatically produce strong short-horizon forecasts during abrupt market shifts. Forecast accuracy would likely improve only with additional information on broader market shocks, refining conditions, or policy disruptions.

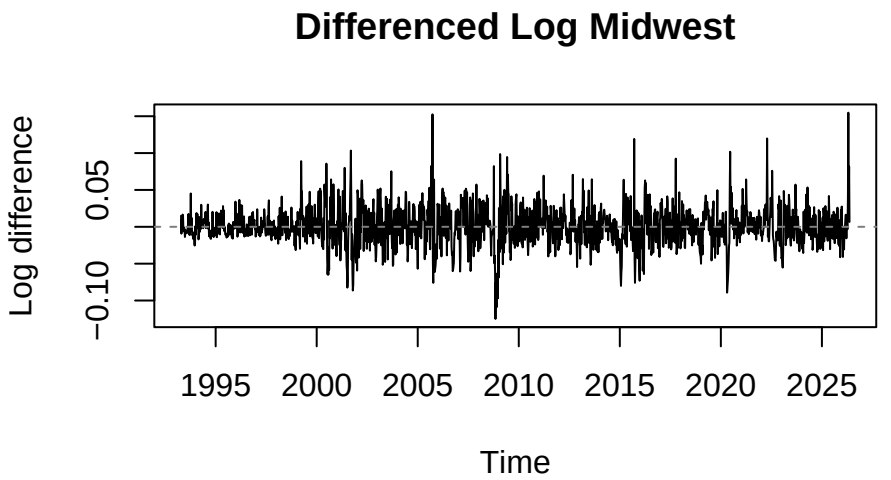
4. Regional U.S. Gasoline Comparison

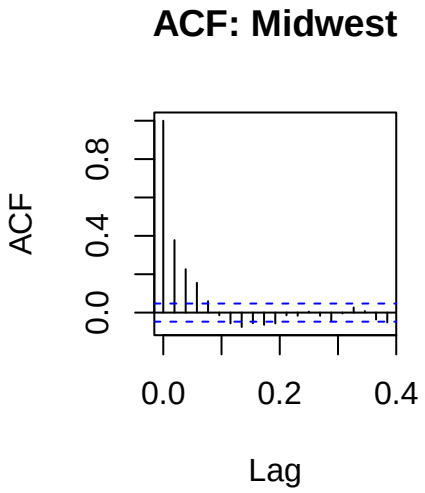
(i) Convert regional series to approximately stationary form

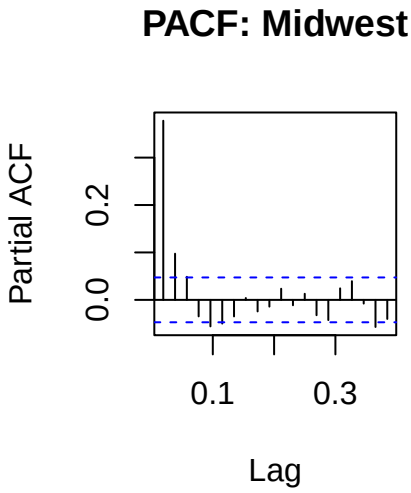
```
# First differences of log prices for all five regions
d_regions <- diff(log(region_var_ts))
colnames(d_regions) <- colnames(region_var_ts)

region_diff_dates <- gas_df$Date[-1]

for (series_name in colnames(d_regions)) {
  plot(d_regions[, series_name],
       main = paste("Differenced Log", gsub("_", " ", series_name)),
       ylab = "Log difference",
       xlab = "Time")
  abline(h = 0, lty = 2, col = "gray50")

  par(mfrow = c(1, 2))
```

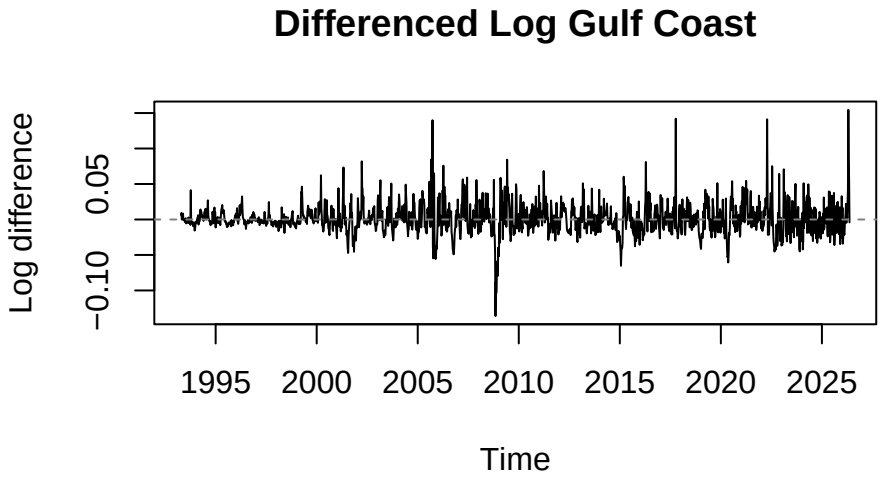
```
  acf(d_regions[, series_name],
      lag.max = 20,
      main = paste("ACF:", gsub("_", " ", series_name)))
  pacf(d_regions[, series_name],
       lag.max = 20,
       main = paste("PACF:", gsub("_", " ", series_name)))
  par(mfrow = c(1, 1))
}
```

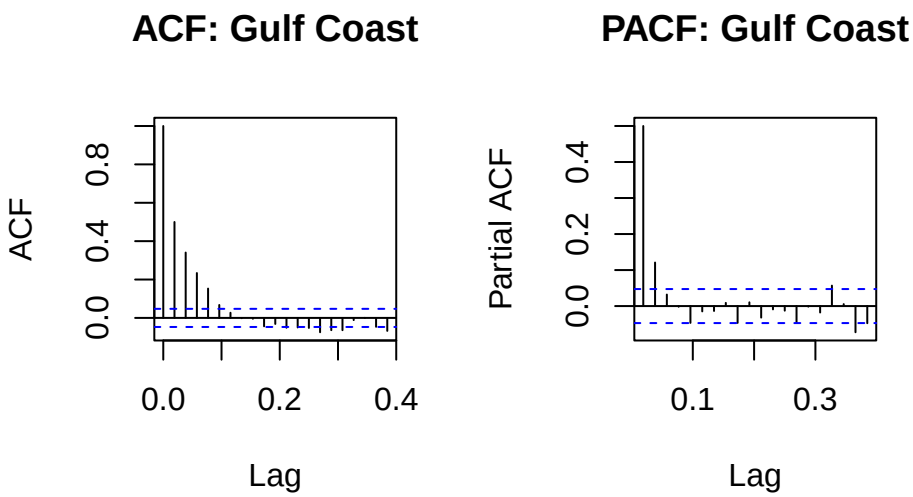


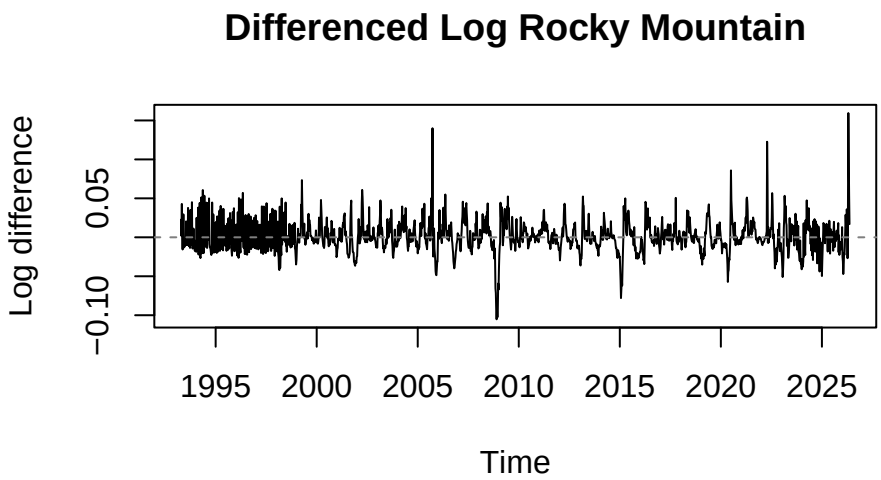

Differenced Log Midwest
Log difference
−0.10
0.05
1995
2000
2005
2010
2015
2020
2025
Time

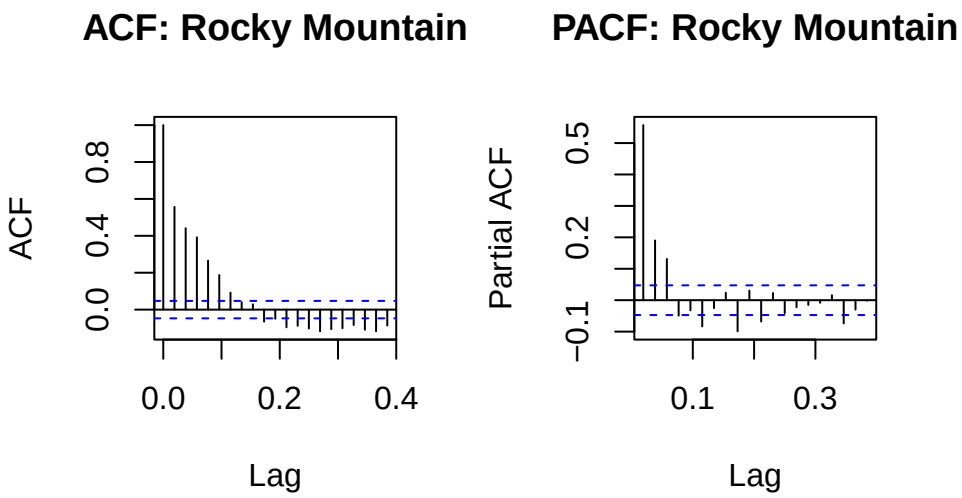

ACF: Midwest
ACF
0.0
0.4
0.8
0.0
0.2
0.4
Lag

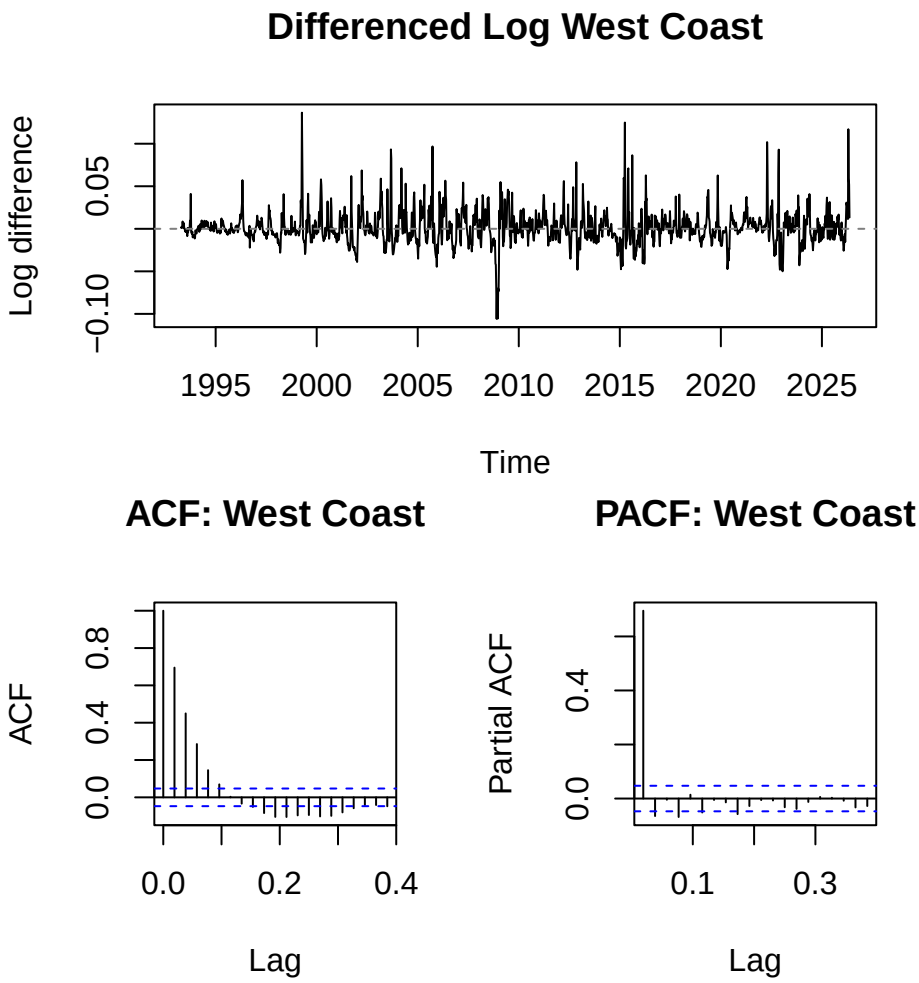

PACF: Midwest
Partial ACF
0.0
0.2
0.1
0.3
Lag


Differenced Log Gulf Coast
Log difference
−0.10
0.05
1995
2000
2005
2010
2015
2020
2025
Time


ACF: Gulf Coast
ACF
0.0
0.4
0.8
0.0
0.2
0.4
Lag
PACF: Gulf Coast
Partial ACF
0.0
0.2
0.4
0.1
0.3
Lag


Differenced Log Rocky Mountain
Log difference
−0.10
0.05
1995
2000
2005
2010
2015
2020
2025
Time


ACF: Rocky Mountain
ACF
0.0
0.4
0.8
0.0
0.2
0.4
Lag
PACF: Rocky Mountain
Partial ACF
−0.1
0.2
0.5
0.1
0.3
Lag

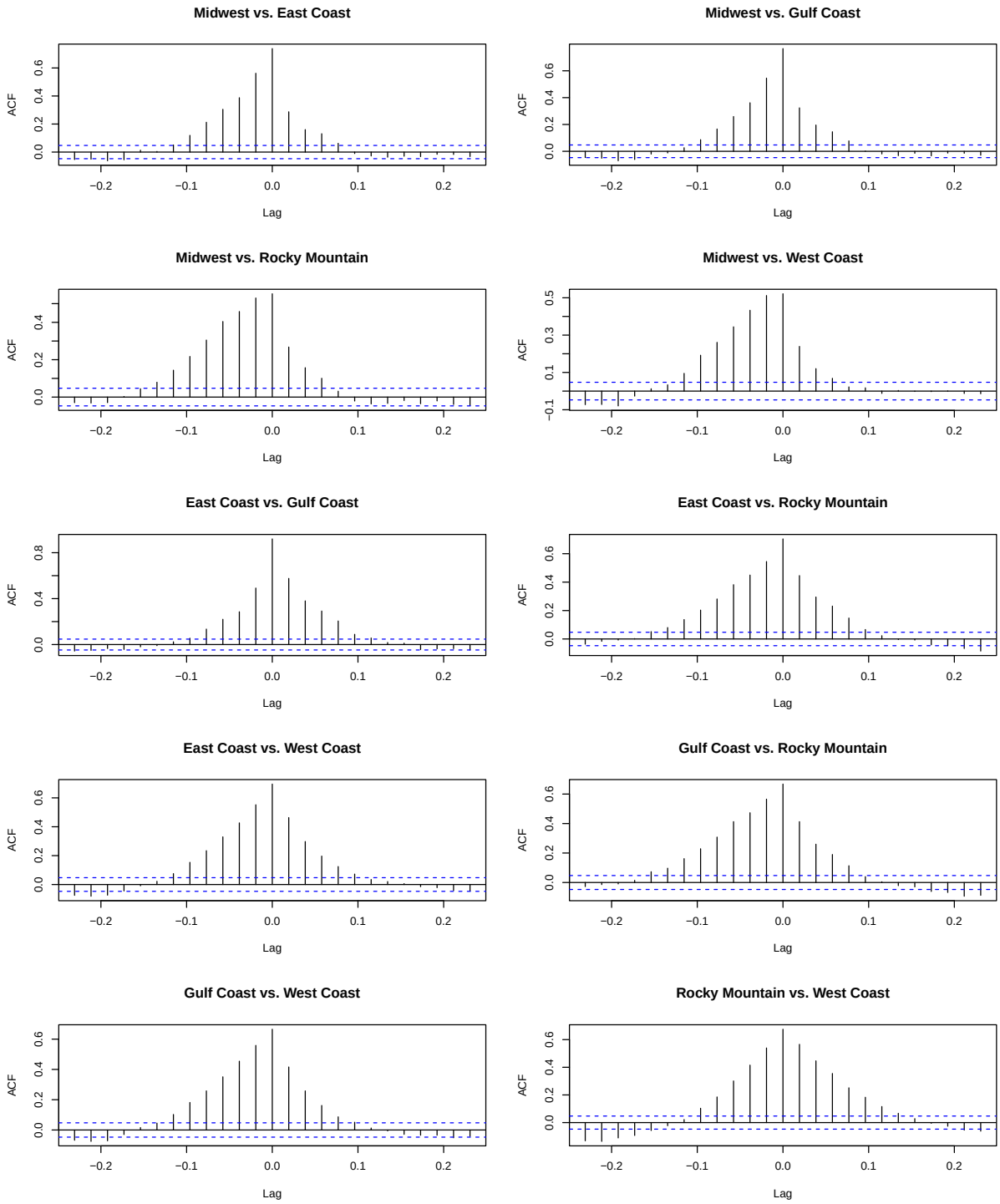


The transformed regional series appear much closer to stationarity than the level series. After differencing and logging, each series fluctuates around zero, the ACFs decay quickly, and there is no strong visual evidence of a remaining trend.

(ii) CCFs for regional lead-lag patterns

```
# Pairwise CCFs for all 10 regional pairs
region_pairs <- combn(colnames(d_regions), 2, simplify = FALSE)

# Show Midwest pairs first because they are especially informative in the plots
region_pairs <- c(
  Filter(function(x) "Midwest" %in% x, region_pairs),
  Filter(function(x) !("Midwest" %in% x), region_pairs)
```

```r
)

region_pairs <- lapply(region_pairs, function(pair) {
  if ("Midwest" %in% pair) {
    c("Midwest", setdiff(pair, "Midwest"))
  } else {
    pair
  }
})

par(mfrow = c(5, 2))

for (pair in region_pairs) {
  ccf(d_regions[, pair[1]],
      d_regions[, pair[2]],
      lag.max = 12,
      main = paste(gsub("_", " ", pair[1]), "vs.", gsub("_", " ", pair[2])))
}
```




```r
par(mfrow = c(1, 1))
```

Most regional pairs have their largest spike at lag 0, which indicates strong shared same-week movement. The Midwest comparisons also show some asymmetry at nearby negative lags, which is consistent with the Midwest sometimes moving slightly earlier than other regions, although the dominant pattern is still contemporaneous co-movement.

(iii) VAR(1) model

```r
region_model_ts <- d_regions

# Candidate lag lengths
VARselect(region_model_ts, lag.max = 8, type = "const")
```

```
$selection
AIC(n)  HQ(n)  SC(n) FPE(n)
     8      3      1      8

$criteria
                   1             2             3             4             5
AIC(n) -4.393438e+01 -4.399915e+01 -4.403971e+01 -4.405313e+01 -4.406199e+01
HQ(n)  -4.389908e+01 -4.393445e+01 -4.394559e+01 -4.392960e+01 -4.390904e+01
SC(n)  -4.383900e+01 -4.382430e+01 -4.378538e+01 -4.371932e+01 -4.364869e+01
FPE(n)  8.308871e-20  7.787730e-20  7.478189e-20  7.378527e-20  7.313512e-20
                   6             7             8
AIC(n) -4.406969e+01 -4.407371e+01 -4.409023e+01
HQ(n)  -4.388733e+01 -4.386194e+01 -4.384905e+01
SC(n)  -4.357691e+01 -4.350145e+01 -4.343849e+01
FPE(n)  7.257477e-20  7.228448e-20  7.110110e-20
```

```r
# Fit the working regional VAR(1)
fit_var_regions <- VAR(region_model_ts, p = 1, type = "const")
summary(fit_var_regions)
```

```
VAR Estimation Results:
=========================
Endogenous variables: East_Coast, Midwest, Gulf_Coast, Rocky_Mountain, West_Coast
Deterministic variables: const
Sample size: 1720
Log Likelihood: 25615.901
Roots of the characteristic polynomial:
```

```
0.6515 0.256 0.256 0.198 0.1704
Call:
VAR(y = region_model_ts, p = 1, type = "const")


Estimation results for equation East_Coast:
===========================================
East_Coast = East_Coast.l1 + Midwest.l1 + Gulf_Coast.l1 + Rocky_Mountain.l1 + West_Coast.l1 +

                   Estimate Std. Error t value Pr(>|t|)
East_Coast.l1     0.0056240  0.0522947   0.108    0.914
Midwest.l1        0.1966443  0.0206092   9.542  < 2e-16 ***
Gulf_Coast.l1     0.2120142  0.0445581   4.758 2.12e-06 ***
Rocky_Mountain.l1 0.0345372  0.0260307   1.327    0.185
West_Coast.l1     0.1164431  0.0274288   4.245 2.30e-05 ***
const             0.0003352  0.0003536   0.948    0.343
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


Residual standard error: 0.01465 on 1714 degrees of freedom
Multiple R-Squared: 0.3784, Adjusted R-squared: 0.3766
F-statistic: 208.7 on 5 and 1714 DF,  p-value: < 2.2e-16


Estimation results for equation Midwest:
========================================
Midwest = East_Coast.l1 + Midwest.l1 + Gulf_Coast.l1 + Rocky_Mountain.l1 + West_Coast.l1 + co

                    Estimate Std. Error t value Pr(>|t|)
East_Coast.l1     -0.3036740  0.0885660  -3.429 0.000621 ***
Midwest.l1         0.3207444  0.0349036   9.189  < 2e-16 ***
Gulf_Coast.l1      0.2389347  0.0754633   3.166 0.001571 **
Rocky_Mountain.l1  0.1179783  0.0440855   2.676 0.007519 **
West_Coast.l1      0.0415396  0.0464533   0.894 0.371328
const              0.0004301  0.0005989   0.718 0.472775
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


Residual standard error: 0.02481 on 1714 degrees of freedom
Multiple R-Squared: 0.1541, Adjusted R-squared: 0.1517
F-statistic: 62.47 on 5 and 1714 DF,  p-value: < 2.2e-16
```

```
Estimation results for equation Gulf_Coast:
===========================================
Gulf_Coast = East_Coast.l1 + Midwest.l1 + Gulf_Coast.l1 + Rocky_Mountain.l1 + West_Coast.l1 +

                   Estimate Std. Error t value Pr(>|t|)
East_Coast.l1     0.0323409  0.0630936   0.513 0.608308
Midwest.l1        0.3041834  0.0248650  12.233  < 2e-16 ***
Gulf_Coast.l1     0.0663471  0.0537594   1.234 0.217317
Rocky_Mountain.l1 0.0660422  0.0314061   2.103 0.035625 *
West_Coast.l1     0.1241504  0.0330929   3.752 0.000182 ***
const             0.0002604  0.0004267   0.610 0.541722
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


Residual standard error: 0.01767 on 1714 degrees of freedom
Multiple R-Squared: 0.327,  Adjusted R-squared: 0.325
F-statistic: 166.6 on 5 and 1714 DF,  p-value: < 2.2e-16


Estimation results for equation Rocky_Mountain:
===============================================
Rocky_Mountain = East_Coast.l1 + Midwest.l1 + Gulf_Coast.l1 + Rocky_Mountain.l1 + West_Coast

                    Estimate Std. Error t value Pr(>|t|)
East_Coast.l1     -0.1804089  0.0555926  -3.245   0.0012 **
Midwest.l1         0.1668014  0.0219089   7.613 4.38e-14 ***
Gulf_Coast.l1      0.2111636  0.0473681   4.458 8.81e-06 ***
Rocky_Mountain.l1  0.2225497  0.0276723   8.042 1.63e-15 ***
West_Coast.l1      0.2856227  0.0291586   9.795  < 2e-16 ***
const              0.0001872  0.0003759   0.498   0.6186
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


Residual standard error: 0.01557 on 1714 degrees of freedom
Multiple R-Squared: 0.428,  Adjusted R-squared: 0.4263
F-statistic: 256.5 on 5 and 1714 DF,  p-value: < 2.2e-16


Estimation results for equation West_Coast:
```

```
=========================================
West_Coast = East_Coast.l1 + Midwest.l1 + Gulf_Coast.l1 + Rocky_Mountain.l1 + West_Coast.l1 +

                    Estimate Std. Error t value Pr(>|t|)
East_Coast.l1     -0.1479101  0.0478854  -3.089  0.00204 **
Midwest.l1         0.1370713  0.0188715   7.263  5.7e-13 ***
Gulf_Coast.l1      0.1048158  0.0408012   2.569  0.01028 *
Rocky_Mountain.l1  0.0684224  0.0238359   2.871  0.00415 **
West_Coast.l1      0.5673976  0.0251161  22.591  < 2e-16 ***
const              0.0002801  0.0003238   0.865  0.38719
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


Residual standard error: 0.01341 on 1714 degrees of freedom
Multiple R-Squared: 0.5184, Adjusted R-squared: 0.517
F-statistic:   369 on 5 and 1714 DF,  p-value: < 2.2e-16



Covariance matrix of residuals:
               East_Coast   Midwest Gulf_Coast Rocky_Mountain West_Coast
East_Coast      0.0002146 0.0002555  0.0002291      1.226e-04  1.092e-04
Midwest         0.0002555 0.0006154  0.0003188      1.786e-04  1.548e-04
Gulf_Coast      0.0002291 0.0003188  0.0003123      1.397e-04  1.276e-04
Rocky_Mountain  0.0001226 0.0001786  0.0001397      2.425e-04  8.887e-05
West_Coast      0.0001092 0.0001548  0.0001276      8.887e-05  1.799e-04

Correlation matrix of residuals:
               East_Coast Midwest Gulf_Coast Rocky_Mountain West_Coast
East_Coast         1.0000  0.7031     0.8850         0.5377     0.5557
Midwest            0.7031  1.0000     0.7271         0.4624     0.4653
Gulf_Coast         0.8850  0.7271     1.0000         0.5075     0.5383
Rocky_Mountain     0.5377  0.4624     0.5075         1.0000     0.4255
West_Coast         0.5557  0.4653     0.5383         0.4255     1.0000
```

```r
roots(fit_var_regions)
```

```
[1] 0.6514917 0.2559523 0.2559523 0.1979514 0.1703924
```

We use a VAR(1) with an intercept as a parsimonious model for the five transformed regional series. The model is stable if all inverse roots lie inside the unit circle, and this specification

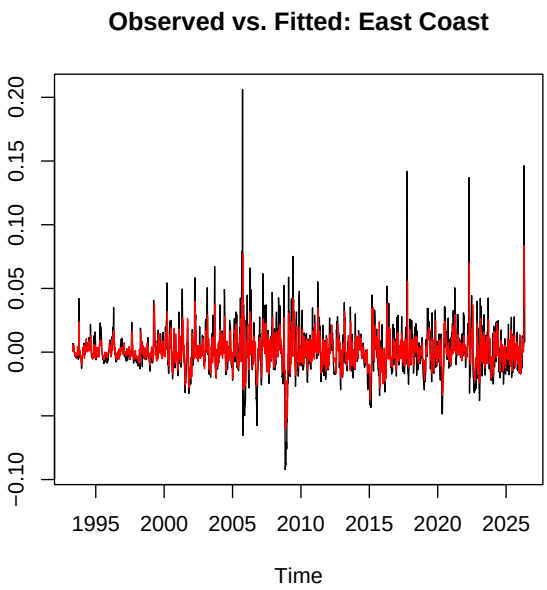
is consistent with the short-memory behavior seen in the univariate ACFs.

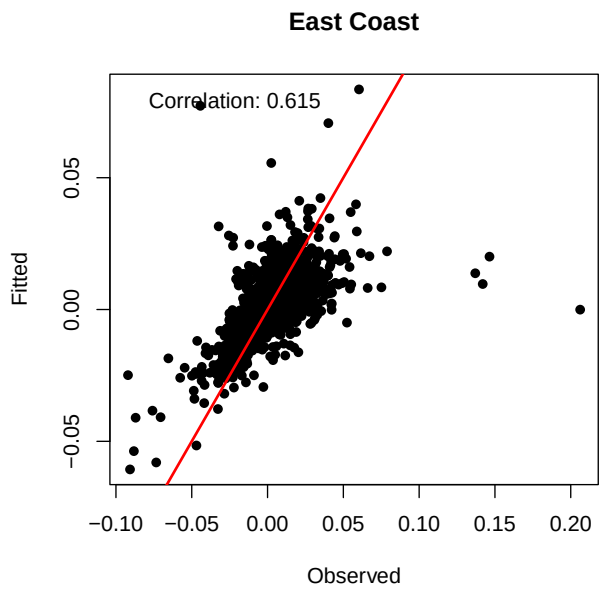
In the fitted equations, the main pattern is that regions do help predict one another, but the strength of the relationships differs across the system. The West Coast and Rocky Mountain series tend to show more distinct regional behavior, while the East Coast, Midwest, and Gulf Coast appear more tightly linked.

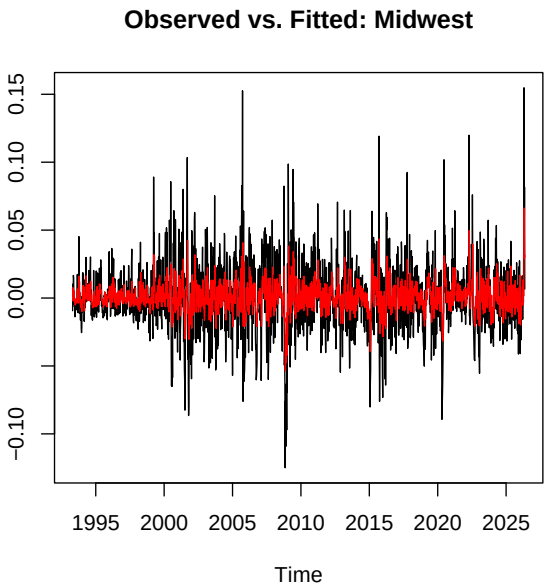
(iv) Diagnostics

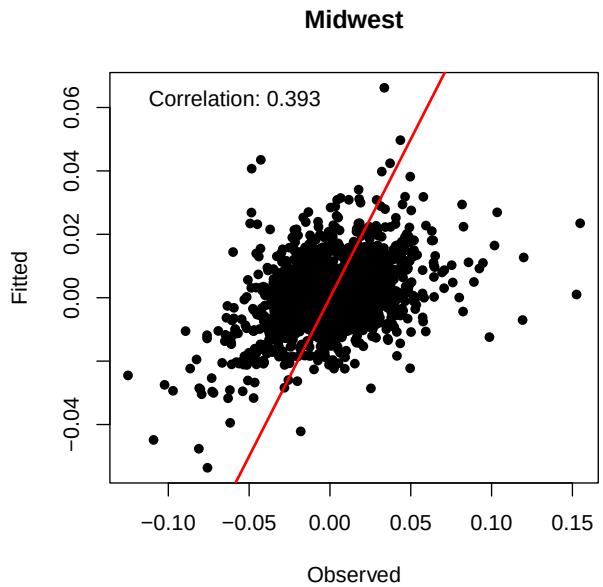
```
# Fitted values and residuals
fitted_regions <- fitted(fit_var_regions)
resid_regions  <- resid(fit_var_regions)
p_used <- fit_var_regions$p
```

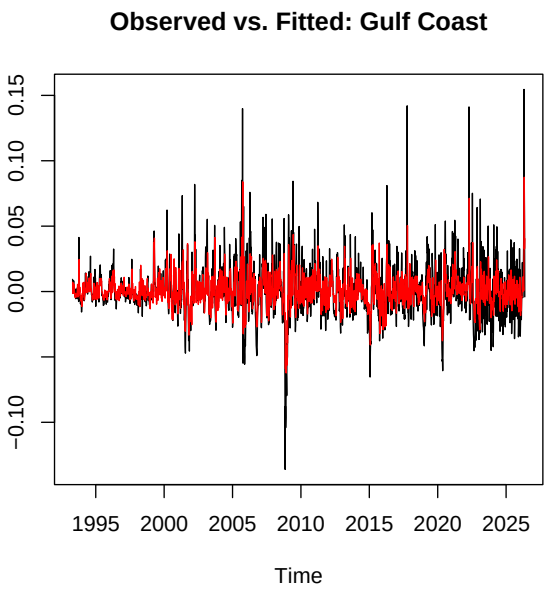
```
# Observed vs. fitted plots for each region
for (series_name in colnames(region_model_ts)) {
  obs_ts <- window(region_model_ts[, series_name], start = time(region_model_ts)[p_used + 1])
  fit_ts <- ts(fitted_regions[, series_name],
               start = start(obs_ts),
               frequency = frequency(obs_ts))

  par(mfrow = c(1, 2))

  plot(obs_ts, type = "l", col = "black",
       main = paste("Observed vs. Fitted:", gsub("_", " ", series_name)),
       ylab = "",
       xlab = "Time")
  lines(fit_ts, col = "red")

  plot(obs_ts, fit_ts,
       pch = 16,
       xlab = "Observed",
       ylab = "Fitted",
       main = gsub("_", " ", series_name))
  abline(0, 1, col = "red", lwd = 2)
  legend("topleft",
         legend = paste("Correlation:", round(cor(obs_ts, fit_ts), 3)),
         bty = "n")

  par(mfrow = c(1, 1))
}
```

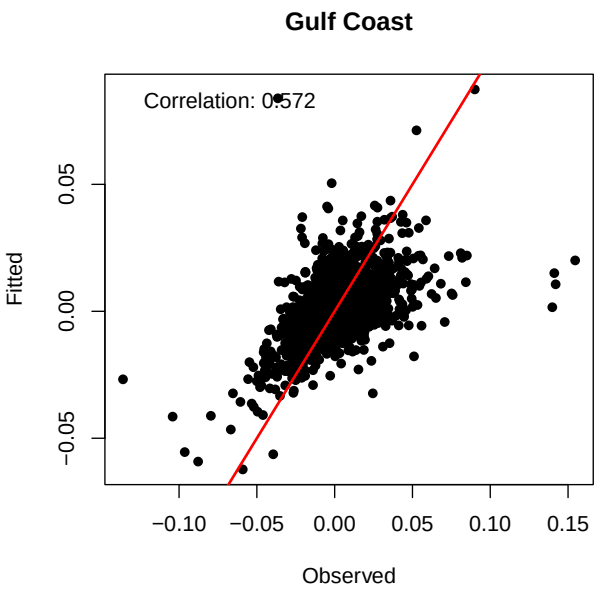

Observed vs. Fitted: East Coast
Time
1995
2000
2005
2010
2015
2020
2025
−0.10
0.00
0.05
0.10
0.15
0.20

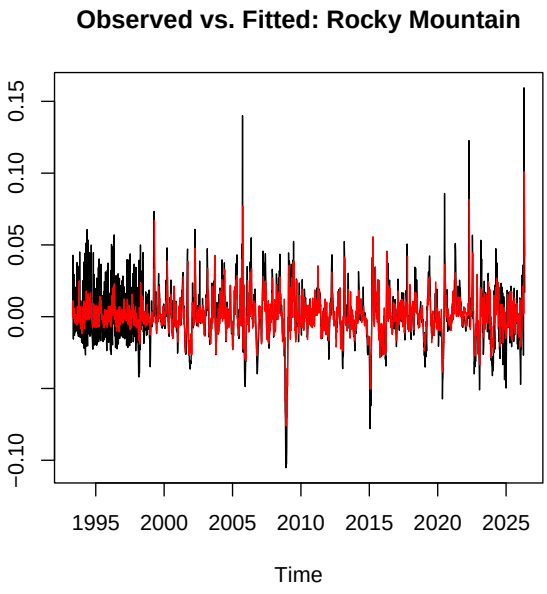

East Coast
Correlation: 0.615
Fitted
Observed
−0.05
0.00
0.05
−0.10
0.10
0.15
0.20

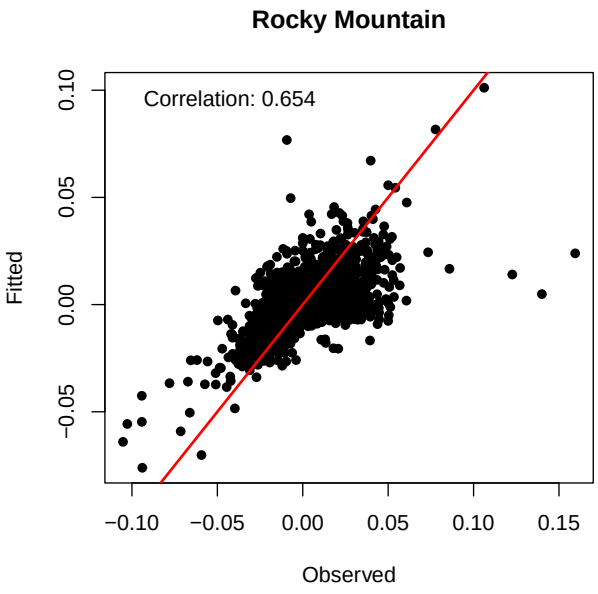

Observed vs. Fitted: Midwest
Time
1995
2000
2005
2010
2015
2020
2025
−0.10
0.00
0.05
0.10
0.15

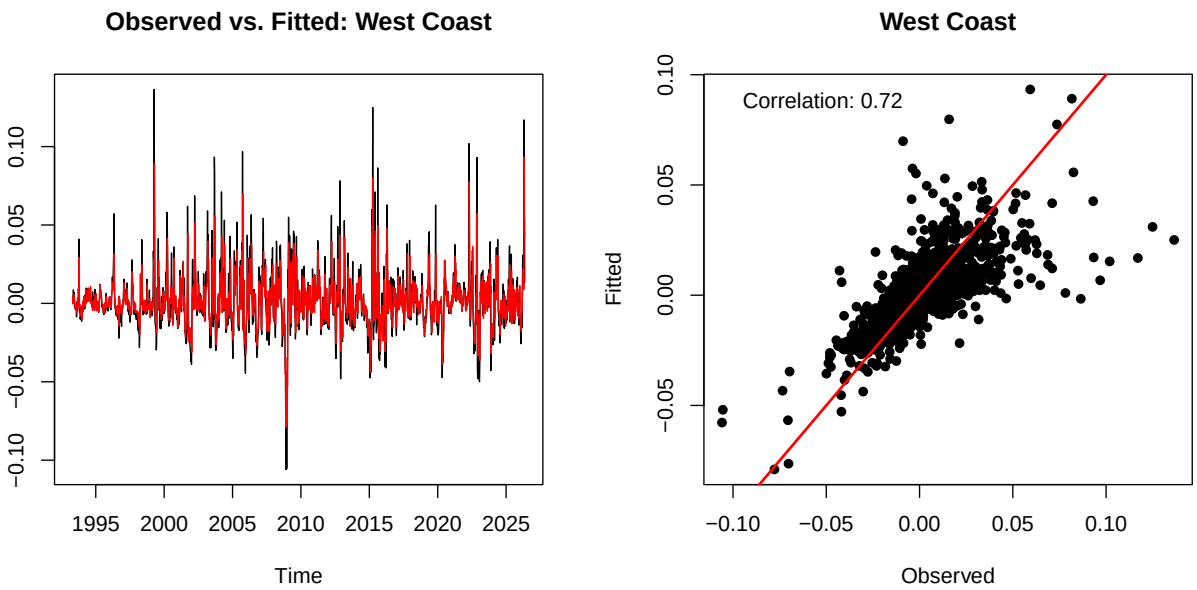

Midwest
Correlation: 0.393
Fitted
Observed
−0.04
0.00
0.02
0.04
0.06
−0.10
−0.05
0.05
0.10
0.15


Observed vs. Fitted: Gulf Coast
0.15
0.10
0.05
0.00
−0.10
1995
2000
2005
2010
2015
2020
2025
Time


Gulf Coast
Correlation: 0.572
Fitted
0.05
0.00
−0.05
−0.10
−0.05
0.00
0.05
0.10
0.15
Observed

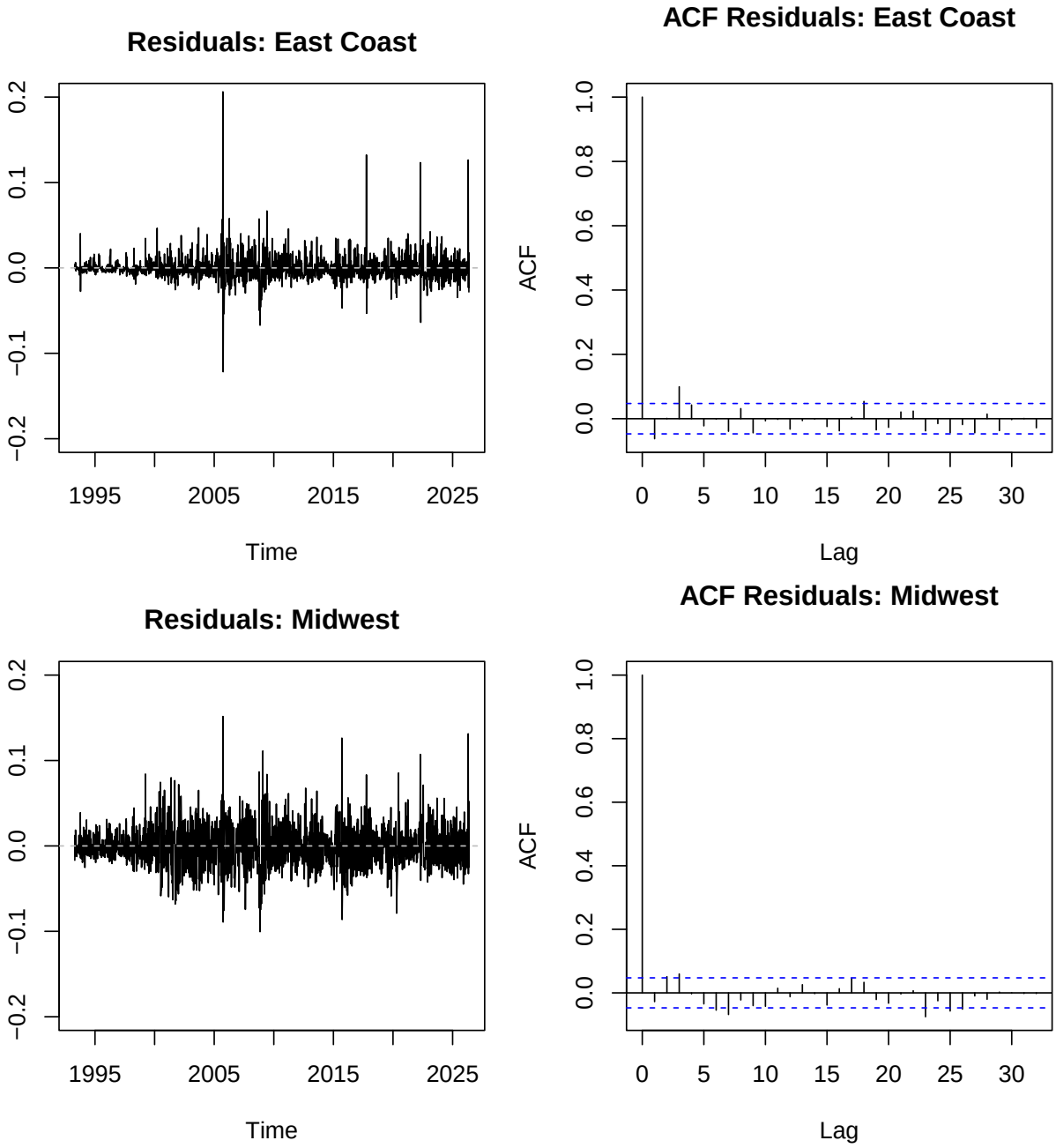

Observed vs. Fitted: Rocky Mountain
0.15
0.10
0.05
0.00
−0.10
1995
2000
2005
2010
2015
2020
2025
Time

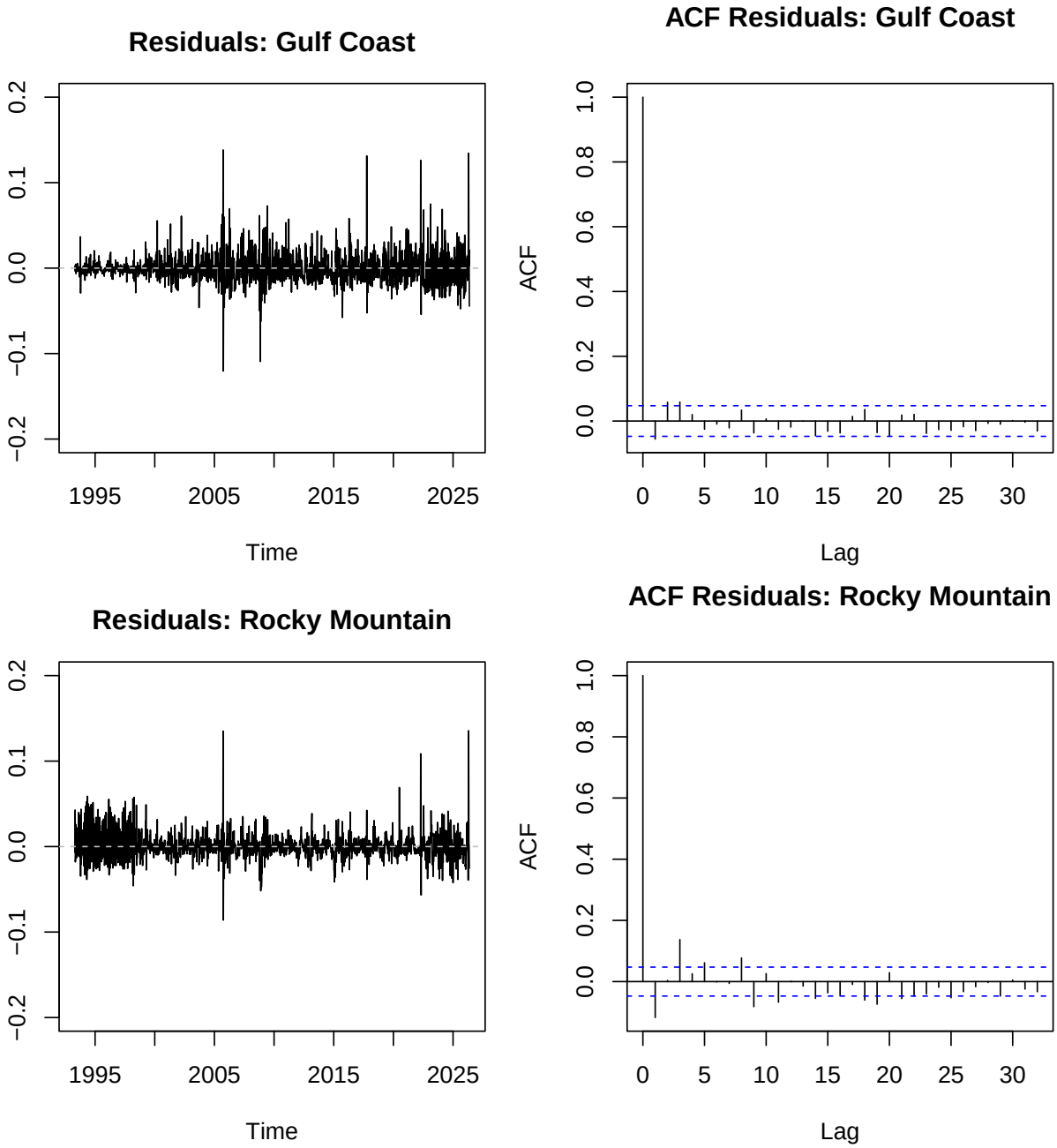

Rocky Mountain
Correlation: 0.654
Fitted
0.10
0.05
0.00
−0.05
−0.10
−0.05
0.00
0.05
0.10
0.15
Observed

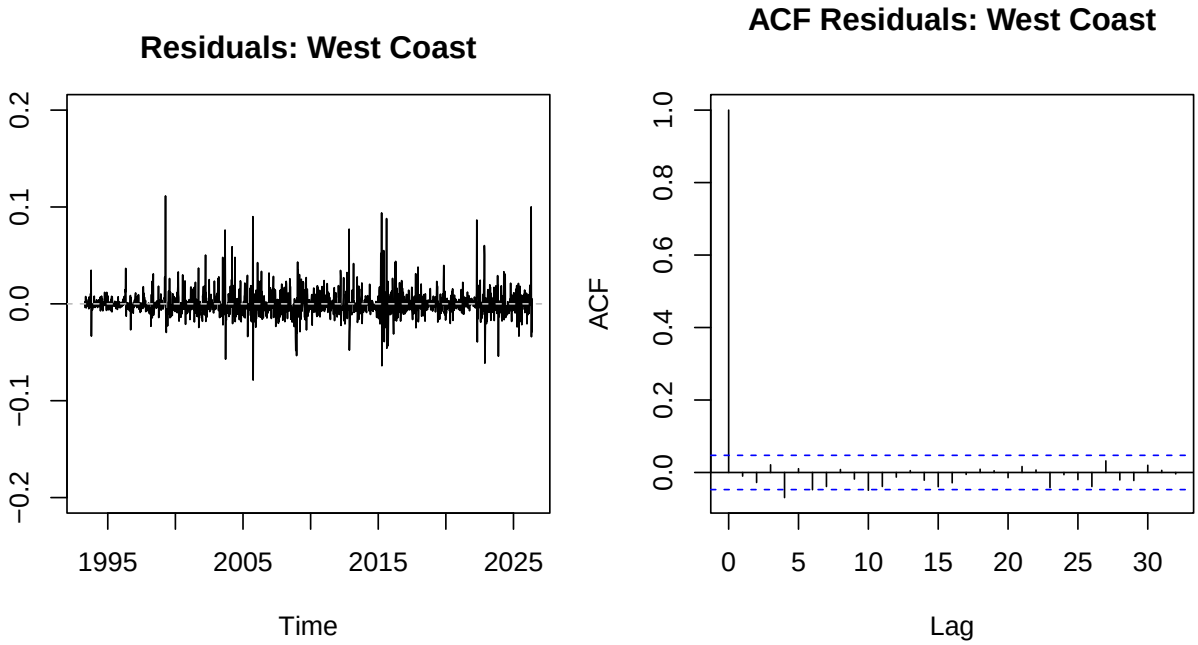


The fitted values capture the center of the regional series reasonably well, but they do not capture abrupt shocks. This is especially noticeable for Rocky Mountain, where the fit appears less stable in earlier periods than in the other regions.

```
# Residual series and ACFs
for (series_name in colnames(resid_regions)) {
  par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))

  plot(ts(resid_regions[, series_name],
          start = time(region_model_ts)[p_used + 1],
          frequency = frequency(region_model_ts)),
       type = "l",
       main = paste("Residuals:", gsub("_", " ", series_name)),
       ylab = "",
       xlab = "Time",
       ylim = c(-0.2, 0.2))
  abline(h = 0, lty = 2, col = "gray")

  acf(resid_regions[, series_name],
      main = paste("ACF Residuals:", gsub("_", " ", series_name)))

  par(mfrow = c(1, 1))
}
```

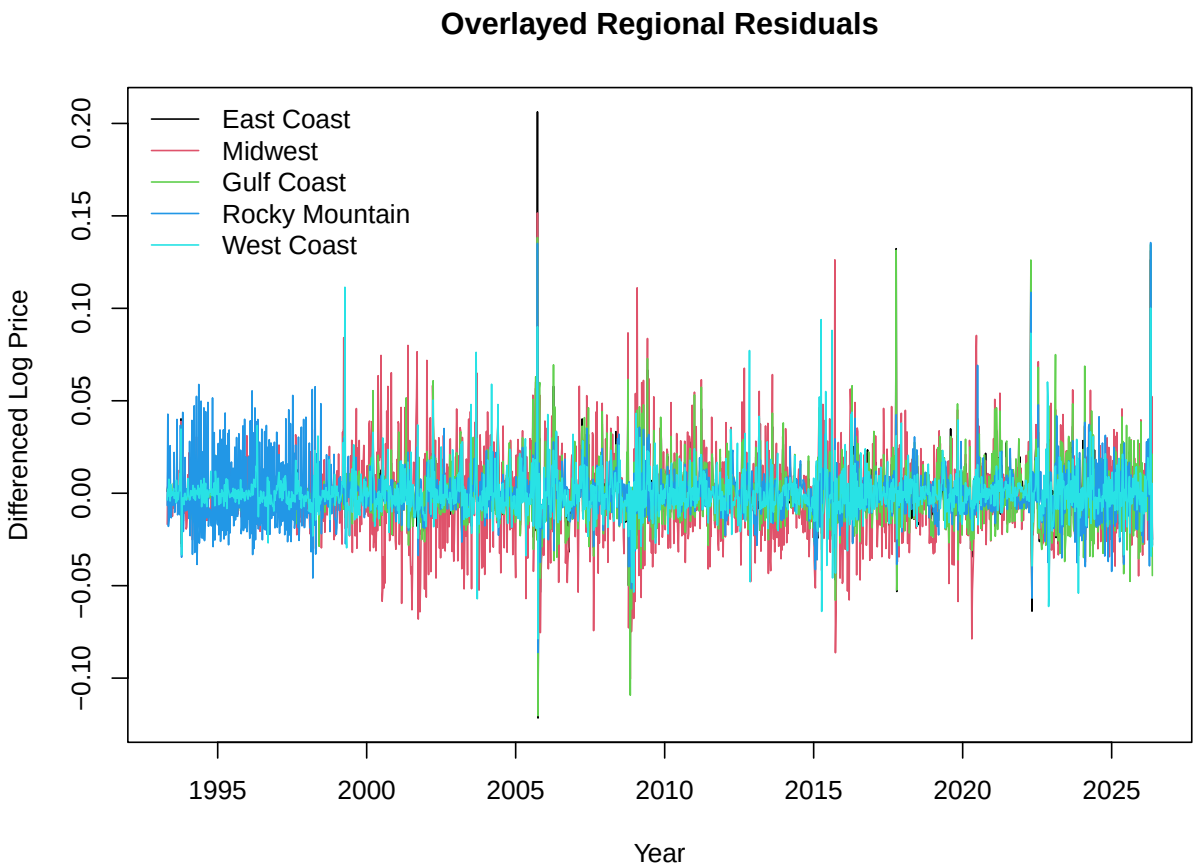

Residuals: East Coast
0.2
0.1
0.0
−0.1
−0.2
1995
2005
2015
2025
Time
ACF Residuals: East Coast
ACF
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25
30
Lag
Residuals: Midwest
0.2
0.1
0.0
−0.1
−0.2
1995
2005
2015
2025
Time
ACF Residuals: Midwest
ACF
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25
30
Lag


Residuals: Gulf Coast
0.2
0.1
0.0
−0.1
−0.2
1995
2005
2015
2025
Time
ACF Residuals: Gulf Coast
ACF
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25
30
Lag
Residuals: Rocky Mountain
0.2
0.1
0.0
−0.1
−0.2
1995
2005
2015
2025
Time
ACF Residuals: Rocky Mountain
ACF
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25
30
Lag



For most regions, the residuals look close to white noise. Rocky Mountain again stands out somewhat, with mild remaining structure relative to the rest of the system.

```
# Overlay residuals from all regional equations
ts.plot(
  ts(resid_regions[, 1], start = time(region_model_ts)[p_used + 1], frequency = frequency(reg
  ts(resid_regions[, 2], start = time(region_model_ts)[p_used + 1], frequency = frequency(reg
  ts(resid_regions[, 3], start = time(region_model_ts)[p_used + 1], frequency = frequency(reg
  ts(resid_regions[, 4], start = time(region_model_ts)[p_used + 1], frequency = frequency(reg
  ts(resid_regions[, 5], start = time(region_model_ts)[p_used + 1], frequency = frequency(reg
  col = 1:5, lty = 1,
  main = "Overlayed Regional Residuals",
  xlab = "Year",
  ylab = "Differenced Log Price"
)
legend(
  "topleft",
  legend = c("East Coast", "Midwest", "Gulf Coast", "Rocky Mountain", "West Coast"),
  col = 1:5, lty = 1, bty = "n"
)
```



This overlay shows that large residuals tend to cluster during the same broad periods, which is consistent with system-wide shocks affecting several regions at once.

```
# Weeks with the largest average and region-specific residuals
region_resid_df <- data.frame(
  time = time(region_model_ts)[-1],
  avg_resid = rowMeans(resid_regions),
  resid_regions
)

region_resid_df |>
  dplyr::select(time, avg_resid) |>
  arrange(desc(avg_resid)) |>
  head(5) |>
  arrange(time)
```

```
         time  avg_resid
647  2005.731 0.14425067
```

```
839  2009.423 0.05523088
1273 2017.769 0.08407856
1508 2022.288 0.11035057
1717 2026.308 0.12557987
```

```r
region_resid_df |>
  dplyr::select(time, East_Coast) |>
  arrange(desc(East_Coast)) |>
  head(5) |>
  arrange(time)
```

```
         time East_Coast
647  2005.731 0.20626972
839  2009.423 0.06671127
1273 2017.769 0.13226865
1508 2022.288 0.12326555
1717 2026.308 0.12634877
```

```r
region_resid_df |>
  dplyr::select(time, Midwest) |>
  arrange(desc(Midwest)) |>
  head(5) |>
  arrange(time)
```

```
         time   Midwest
647  2005.731 0.1515660
821  2009.077 0.1110836
1166 2015.712 0.1262244
1508 2022.288 0.1071650
1717 2026.308 0.1313150
```

```r
region_resid_df |>
  dplyr::select(time, Gulf_Coast) |>
  arrange(desc(Gulf_Coast)) |>
  head(5) |>
  arrange(time)
```

```
         time Gulf_Coast
647  2005.731 0.13828130
1273 2017.769 0.13138052
```

```
1508 2022.288 0.12613670
1551 2023.115 0.07496655
1717 2026.308 0.13452730
```

```r
region_resid_df |>
  dplyr::select(time, Rocky_Mountain) |>
  arrange(desc(Rocky_Mountain)) |>
  head(5) |>
  arrange(time)
```

```
         time Rocky_Mountain
56   1994.365     0.05882623
647  2005.731     0.13509356
1415 2020.500     0.06918833
1508 2022.288     0.10862619
1717 2026.308     0.13556882
```

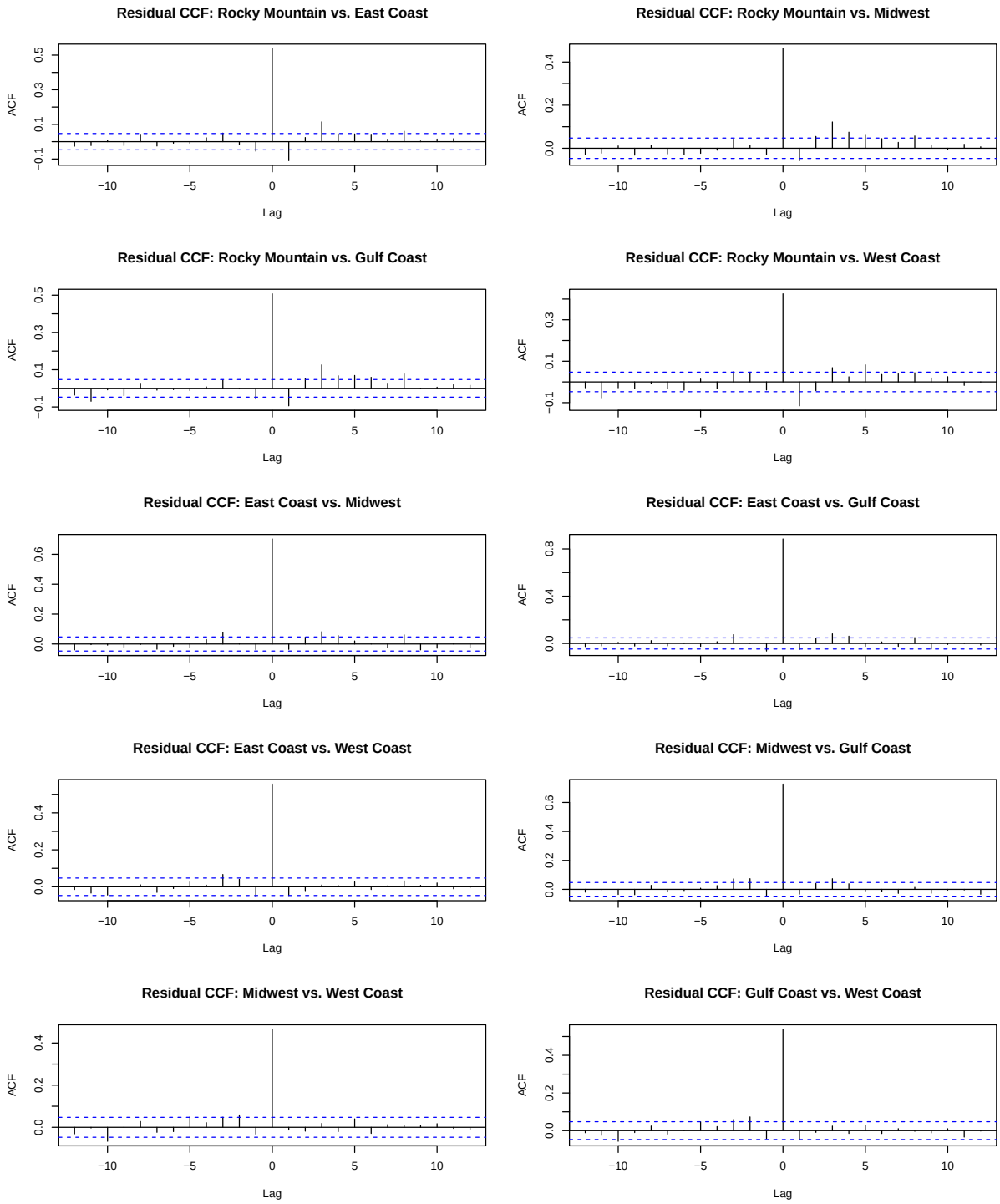
```r
region_resid_df |>
  dplyr::select(time, West_Coast) |>
  arrange(desc(West_Coast)) |>
  head(5) |>
  arrange(time)
```

```
         time West_Coast
311  1999.269 0.11140799
647  2005.731 0.09004273
1142 2015.250 0.09389421
1161 2015.615 0.08805316
1717 2026.308 0.10013947
```

The largest residual weeks overlap across several regions, which again suggests that the biggest misses are driven by common market shocks rather than by isolated local noise.

```r
# Residual CCFs for all regional pairs
region_resid_pairs <- combn(colnames(resid_regions), 2, simplify = FALSE)

region_resid_pairs <- c(
  Filter(function(x) "Rocky_Mountain" %in% x, region_resid_pairs),
  Filter(function(x) !("Rocky_Mountain" %in% x), region_resid_pairs)
)
```

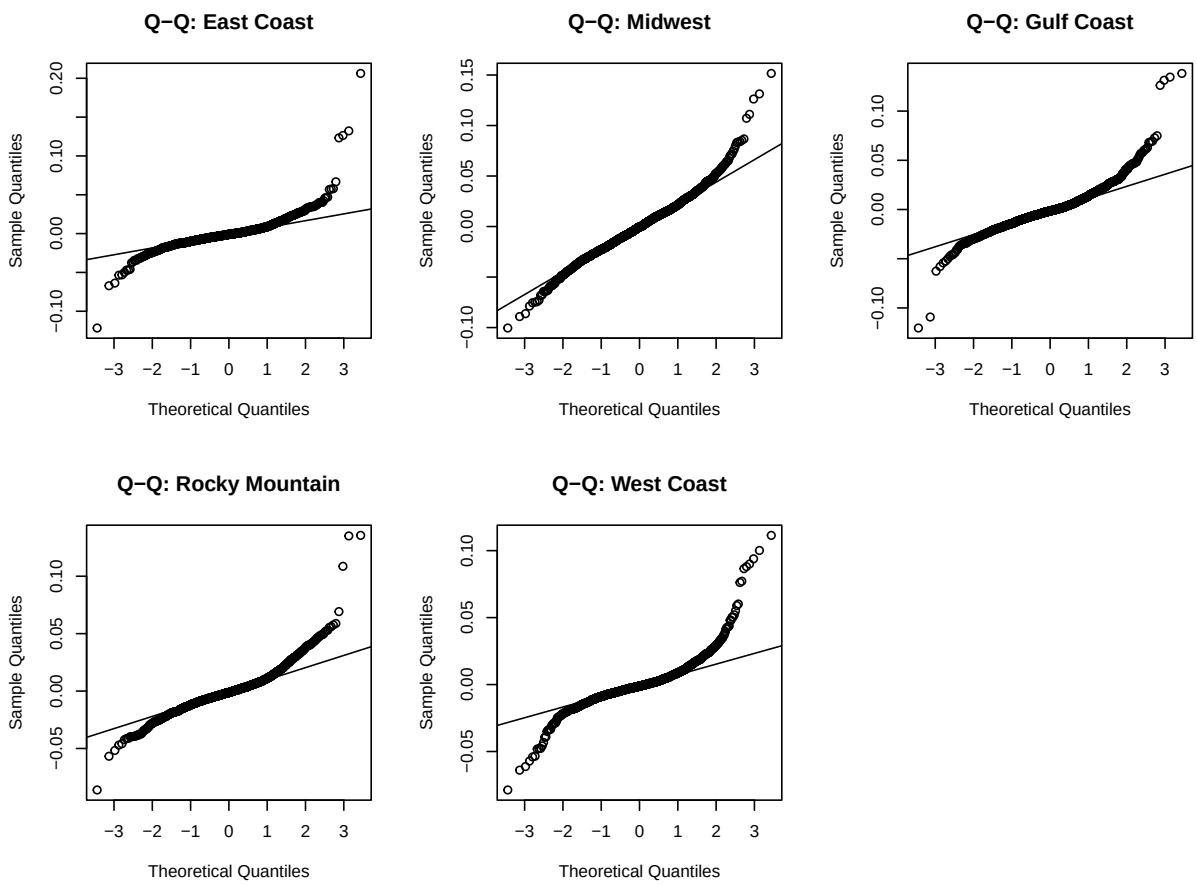
```r
region_resid_pairs <- lapply(region_resid_pairs, function(pair) {
  if ("Rocky_Mountain" %in% pair) {
    c("Rocky_Mountain", setdiff(pair, "Rocky_Mountain"))
  } else {
    pair
  }
})

par(mfrow = c(5, 2))

for (pair in region_resid_pairs) {
  ccf(resid_regions[, pair[1]],
      resid_regions[, pair[2]],
      lag.max = 12,
      main = paste("Residual CCF:",
                   gsub("_", " ", pair[1]), "vs.", gsub("_", " ", pair[2])))
}
```




```r
par(mfrow = c(1, 1))
```

For most pairs, the residual cross-correlations are concentrated at lag 0, which suggests the VAR(1) has removed most of the lagged dependence. Rocky Mountain still looks a bit less regular than the other regions, which is consistent with the earlier diagnostic plots.

```
# Normal Q-Q plots
par(mfrow = c(2, 3))

for (series_name in colnames(resid_regions)) {
  qqnorm(resid_regions[, series_name],
         main = paste("Q-Q:", gsub("_", " ", series_name)))
  qqline(resid_regions[, series_name])
}

par(mfrow = c(1, 1))
```



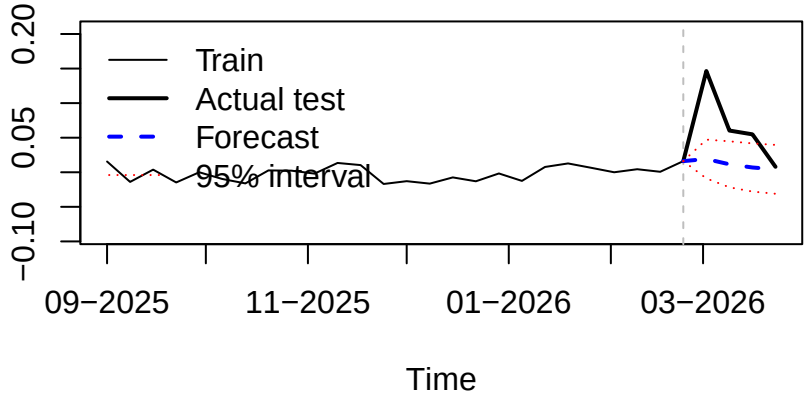
The Q-Q plots show heavier tails than a normal distribution, which is not surprising for weekly energy price changes. The main message is that the regional VAR captures ordinary week-to-week dynamics better than it captures large shock periods.

(v) Forecast

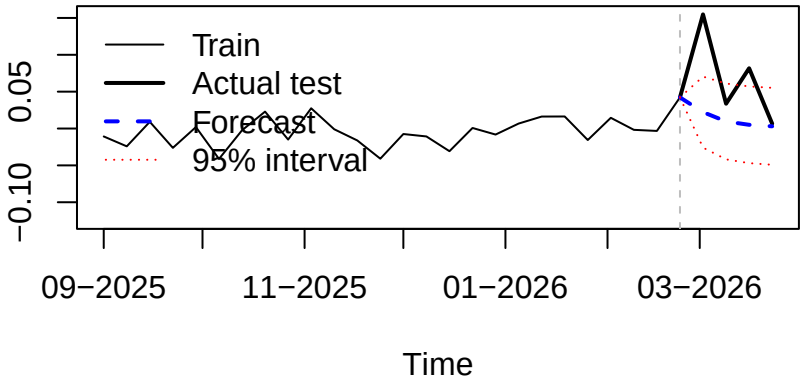
```
# Hold out the last 4 weeks
forecast_horizon <- 4

region_train_ts <- region_model_ts[1:(nrow(region_model_ts) - forecast_horizon), ]
region_test_ts  <- region_model_ts[(nrow(region_model_ts) - forecast_horizon + 1):nrow(regio

region_train_dates <- region_diff_dates[1:(length(region_diff_dates) - forecast_horizon)]
region_test_dates  <- region_diff_dates[(length(region_diff_dates) - forecast_horizon + 1):le

fit_var_regions_train <- VAR(region_train_ts, p = 1, type = "const")
region_forecast <- predict(fit_var_regions_train, n.ahead = forecast_horizon, ci = 0.95)

for (series_name in colnames(region_train_ts)) {
  y_train <- region_train_ts[, series_name]
  y_test  <- region_test_ts[, series_name]
  fc_df   <- as.data.frame(region_forecast$fcst[[series_name]])

  plot_start_idx <- max(1, length(region_train_dates) - 26 + 1)
  recent_dates <- region_train_dates[plot_start_idx:length(region_train_dates)]
  recent_train_values <- y_train[plot_start_idx:length(y_train)]

  plot(
    x = recent_dates,
    y = recent_train_values,
    type = "l",
    xlim = c(min(recent_dates), max(region_test_dates)),
    ylim = range(c(y_train, y_test, fc_df$fcst, fc_df$lower, fc_df$upper), na.rm = TRUE),
    xlab = "Time",
    ylab = "",
    main = paste("Actual vs. Forecast:", gsub("_", " ", series_name), "(4-Week Test Set)"),
    xaxt = "n"
  )

  tick_pos <- seq(from = min(recent_dates), to = max(region_test_dates), by = "1 month")
  axis(1, at = tick_pos, labels = format(tick_pos, "%m-%Y"))

  abline(v = tail(region_train_dates, 1), lty = 2, col = "gray")

  lines(
    x = c(tail(region_train_dates, 1), region_test_dates),
```

```r
    y = c(tail(y_train, 1), y_test),
    col = "black",
    lwd = 2
  )

  lines(
    x = c(tail(region_train_dates, 1), region_test_dates),
    y = c(tail(y_train, 1), fc_df$fcst),
    col = "blue",
    lty = 2,
    lwd = 2
  )

  lines(
    x = c(tail(region_train_dates, 1), region_test_dates),
    y = c(tail(y_train, 1), fc_df$lower),
    col = "red",
    lty = 3
  )

  lines(
    x = c(tail(region_train_dates, 1), region_test_dates),
    y = c(tail(y_train, 1), fc_df$upper),
    col = "red",
    lty = 3
  )

  legend(
    "topleft",
    legend = c("Train", "Actual test", "Forecast", "95% interval"),
    col = c("black", "black", "blue", "red"),
    lty = c(1, 1, 2, 3),
    lwd = c(1, 2, 2, 1),
    bty = "n"
  )
}
```

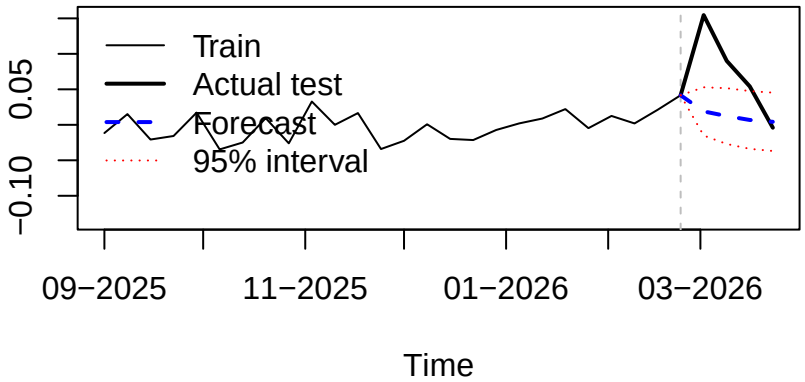
Actual vs. Forecast: East Coast (4−Week Test Set)

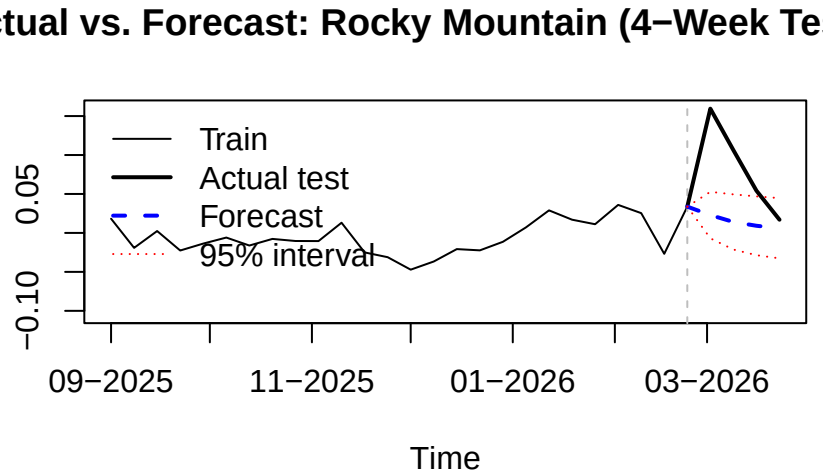


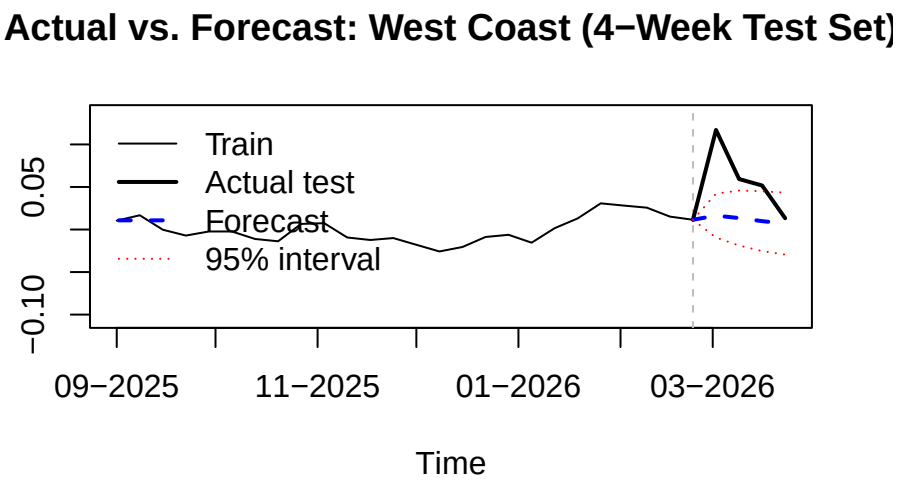
Actual vs. Forecast: Midwest (4−Week Test Set)



Actual vs. Forecast: Gulf Coast (4−Week Test Set)







Across regions, the 4-week forecasts again capture the average direction better than the exact week-to-week movements. Some regions, especially East Coast and West Coast, appear to show at least part of the short rise before dropping back, while others flatten out too quickly.

This reinforces the same point seen earlier: the regional VAR is useful for describing dependence across markets, but it is still limited as a short-horizon forecasting tool when the system is hit by sudden shocks.